DialectLLM: A Dialect-Aware Dialog[ue] Generation Framework Beyond Standard American English

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Abstract

More than 80% of the 1.6B English speakers do not use Standard American English (SAE), yet LLMs often fail to correctly identify non-SAE dialects and generate stereotyped responses for their speakers. We introduce **DialectLLM**, the first large-scale framework for generating high-quality multi-dialectal conversational data encompassing the three pillars of written dialect—lexical (vocabulary), orthographic (spelling), and morphosyntactic (grammar) features. DialectLLM produces a dialect-parallel dialog dataset spanning nine English dialects. Partnering with native linguists, we design and validate SAE-to-dialect transformation rules, ensuring authenticity. Our approach challenges the prevailing practice of applying a single morphosyntactic feature set to both user utterances and model responses, showing that models should not reproduce up to 90% of the grammatical features of a dialect. Human evaluation confirms data quality, with annotators preferring DialectLLM over prior methods in 98.8% of pairwise comparisons for dialect naturalness. We then construct **DialectLLM-Bench**, a dialect-parallel benchmark with 50k+ dialogs, resulting in 97k+ QA pairs, and evaluate 17 LLMs on dialect identification and response generation tasks. Even frontier models achieve under 70% accuracy, fail to reach 50% for prominent dialects like Canadian English, and systematically misclassify non-SAE dialects as American or British. Beyond benchmarking, we show that DialectLLM data also serve as a scalable LLM post-training resource, suggesting a practical path toward dialect-aware conversational AI.

1 Introduction

Large Language Models (LLMs) are increasingly deployed in conversational AI, yet their training data skew heavily toward Standard American English (SAE). The limited dialect-specific data available are often narrow in domain and exaggerated in form, causing models to learn stereotypical and potentially offensive patterns [1–3]. These deficiencies persist beyond post-training: models exhibit degraded performance for non-SAE users across classification, question-answering, and reasoning tasks [4–12]. More troublingly, LLMs have been shown to display explicit bias against users of non-standard dialects [3, 13, 14]. These deficits not only degrade user satisfaction [15], but also disadvantage dialect speakers in real-world LLM applications [10].

To narrow this gap, various dialect-specific data generation methods have been proposed. Prior approaches typically transform SAE text into a target dialect using human translation [10], rule-based heuristics [4], LLMs [11], or hybrid methods [12]. LLMs offer scalability, but relying on parametric knowledge alone proves insufficient: even state-of-the-art models struggle to produce natural-sounding dialect text [14, 16, 17]. As a result, recent works mostly rely on structured external linguistic resources such as the Electronic World Atlas of Varieties of English (eWAVE) to apply rule-guided transformations.

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Table 1: Example Irish dialect transformation comparing eWAVE LLM-based generation [12] and DialectLLM. Highlighted spans indicate transformation type: lexical substitutions, orthographic changes, and morphosyntactic edits. Note the lack of lexical or orthographic changes in prior methods, as well as the thou transformation, which our Irish annotators flagged as archaic.

SAE baseline	Trans-EnV (eWAVE + LLM)	Ours (DialectLLM-Model)
I'm trying to organize my linen closet better, but it's such a mess right now. Do you have any tips for keeping sheets neat and accessible?	I'm trying to organize me linen closet where it's, like, such a mess right now more good. Thou have any tips for keeping sheets neat and accessible?	I'm trying to organise me hot press better, but it's such a mess right now. What tips ye have for keeping sheets neat and accessible?

However, our analysis of previous work and eWAVE itself reveals three critical limitations. First, eWAVE indexes only morphosyntactic variation, omitting orthographic variations (e.g., *color* vs. *colour*) and lexical variations (e.g., *apartment* vs. *flat*), which are central to written dialect. Second, evaluation with native speakers reveals that direct application of eWAVE rules produces exaggerated and often antiquated representations of dialects. Third—and most critically—prior work conflates dialect features used by speakers with those appropriate for model generation. For example, the discourse marker *like* ("Carol was, like, making lunch") is pervasive in spoken English, including SAE, yet is implicitly understood to be inappropriate for generated written text. By treating all attested eWAVE features as suitable for model output, dialect-specific generators risk reinforcing the very stereotypes they aim to correct.

To address these gaps, we introduce DialectLLM, a comprehensive framework for generating multi-turn, dialect-parallel dialogs. We partner with native linguists to identify the key lexical (vocabulary; parking lot → car park), orthographic (spelling; color → colour), and morphosyntactic (grammar; Sarah and I → Sarah and meself) differences between SAE and their native dialects [18]. We also ask them to categorize these features as appropriate or inappropriate for AI-generated model responses. Annotators indicate that up to 90% of the eWAVE features should not be replicated in model responses, depending on the dialect.

Using SoTA LLMs, we generate seed dialogs and apply the rule-guided transformations identified as appropriate by native linguists. After a series of post-transformation quality checks, DialectLLM creates a collection of dialect-accurate multi-turn conversations alternating between user requests and model responses. In total, we produce 3,680 dialogs parallel across 9 English dialects–American (SAE; US), Australian (AU), British (GB), Canadian (CA), Indian (IN), Irish (IE), Nigerian (NG), Philippine (PH), and Scottish (SC)–with four turn variants (1, 2, 4, and 8) and three different datasets, resulting in a total of 400k+ generated dialogs.

Using this data, we construct DialectLLM-Bench, spanning 9 English dialect variants to benchmark LLM dialect classification and response generation capabilities. Although dialect identification is a fundamental prerequisite to dialect natural language understanding (NLU) and appropriate responses, little research has examined models' capacity to discern English dialects [19]. We present LLMs with our generated data and ask how well the model can a) identify the dialect and b) select the dialect-appropriate model response. Probing 17 LLMs across diverse families and sizes, we observe that even SoTA LLMs achieve less than 70% accuracy on average on both classification and generation tasks. These deficiencies are even more acute for certain dialects, as all tested models do not reach 40% accuracy for CA. Examining failure patterns, we see a strong bias towards US (for CA & PH) or GB (for IE, SC, & IN) dialects. Collectively, this work suggests that dialect misidentification forms as an early failure point that can propagate to many observed non-SAE defects in tasks such as question-answering or reasoning.

We further demonstrate that post-training on DialectLLM data substantially improves dialectal capabilities in LLMs, even with limited supervision. Notably, fine-tuning for dialectal response generation induces strong gains in dialect identification, suggesting that models acquire internal representations of systematic variation.

In summary, (1) we propose **DialectLLM**, a novel dialect-aware data generation pipeline along lexical, orthographic, and morphosyntactic dimensions, using LLM-based transformations with native linguist annotations. (2) We explicitly delineate between morphosyntactic features common in

users' dialects from those the model should use when generating text. (3) We construct a large-scale parallel multi-dialectal conversational dataset spanning 9 English dialects, enabling future research on dialect-aware conversational AI. (4) We demonstrate the utility of DialectLLM data for both benchmarking and post-training: **DialectLLM-Bench** shows the current LLMs' inability on dialect identification and generation, while post-training on DialectLLM data improves dialectal capabilities.

2 Related Work

Dialect boundaries are inherently fluid, but in this paper we define dialects at the country level. Although regional and sub-regional differences exist within countries, this high-level categorization captures meaningful variation across the world. Given a dearth of high-quality dialect-specific data, particularly for lower-resource English dialects, substantial effort has been made to transform SAE text into target dialects. Hiring native speakers to manually transform text into their local dialect is effective but financially prohibitive to scale [10, 20]. As a result, most techniques transform baseline text by (a) using LLMs' parametric knowledge, (b) applying rule guided transformations (RGTs), or a combination of the two.

Parametric transformations are the simplest and commonly used [11, 21], but risk reproducing the very biases that these data aim to correct [1]. Instead, most work applies RGTs with deterministic syntax parsing rules or an LLM. While some work targets lexical differences [22], most research exclusively focuses on the morphosyntactic features of a dialect, using eWAVE, a database of 235 morphosyntactic features used across 77 English varieties, compiled by professional linguists from 175 peer-reviewed sources [23], as a source of ground truth rules [4, 7, 12, 20, 24]. Its appeal is clear: comprehensive coverage, scientific grounding, and public availability.

Prior works [4, 20] apply eWAVE using rule-based heuristics without LLMs, but these methods fail to capture the contextual depth of real-world dialect usages [10] and often produce exaggerated feature applications (see App. C.3 for examples). Gupta et al. [11] prompt GPT-4o with three eWAVE few-shot examples to transform SAE text to different dialects. Lee et al. [12] provide GPT-4o-mini with eWAVE rules and LLM-generated guidelines alongside SAE text, using eWAVE's attestation ratings to assist with transformations. DialectLLM adopts a hybrid approach by using LLMs to apply native linguist validated lexical, orthographic, and morphosyntactic transformations, while uniquely distinguishing between features for model comprehension versus generation.

3 DialectLLM

We propose DialectLLM, a novel data generation framework to create a parallel conversational dataset across nine dialects. To properly integrate parametric knowledge from LLMs, rules from linguistic databases, and human-in-the-loop validation from native linguists, we decompose the transformation into five main components. This decomposition prevents LLMs from collapsing to SAE, which is often observed in modern LLMs [1]: (1) identify candidate transformations for SAE $\rightarrow$ target dialect for lexical, orthographic, and morphosyntactic dimensions, (2) validate these transformations with native linguists, (3) construct seed dialogs in SAE, (4) use LLMs to transform the seed dialogs via explicit transformation rules, and finally (5) apply quality controls to ensure accurate transformation. We visualize and formalize the entire generation process in Fig. 1 and Algorithm 1, respectively.

3.1 Dialect Knowledge Base Construction

The first stage of DialectLLM is to build a set of valid transformations from SAE to a target dialect. This consists of two sub-steps: (a) generating candidate transformations and (b) winnowing valid transformations from the candidate list to build the final set of transformations.

Identifying Candidate Transformations. Following prior research, we collect our candidate morphosyntactic transformations from eWAVE [4–7, 12, 20]. Unfortunately, no similar comprehensive database exists to inform lexical and orthographic transformations for English dialects. To collect these additional dimensions, we first prompt GPT-5 and Gemini-3 with web search enabled to collect candidate mappings for each dialect. Then, with Claude-4-Sonnet as a verification filter, we validate each candidate as a genuine difference.

Identifying Valid Transformations with Human Annotators. Critically, relying solely on LLM-based mappings may reinforce the very biases DialectLLM seeks to correct, hence we recruit native linguists to validate the authenticity of each feature transformation. Where possible, we recruit

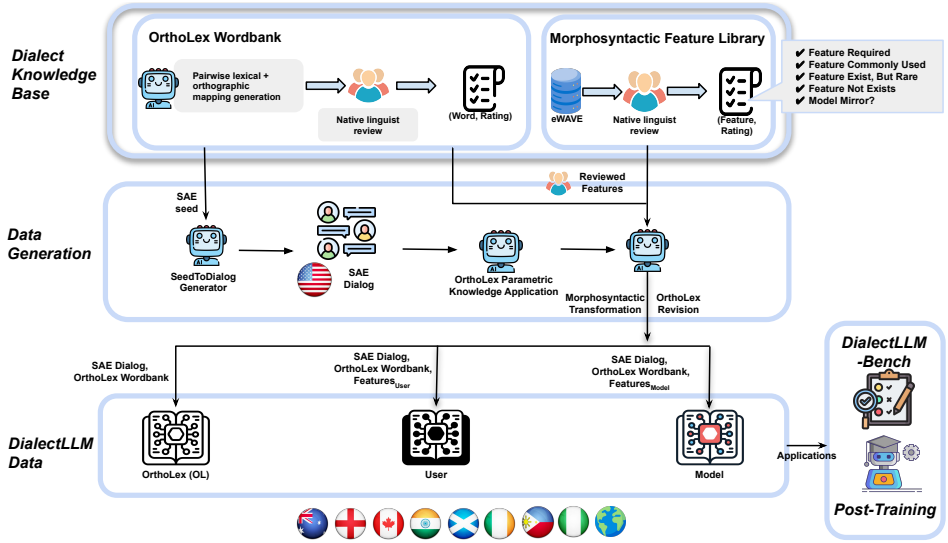


Figure 1: *Overview of DialectLLM.* The framework combines ortho-lexical seed creation, multi-turn SAE dialog generation, and rule-guided transformation using linguist-curated lexical, orthographic, and morphosyntactic knowledge to produce parallel dialogs across 9 English dialects. The resulting dataset supports both benchmarking and post-training to enhance LLMs' dialectal capabilities.

dialect-expert linguists; otherwise, we recruit three annotators from diverse backgrounds per dialect (e.g., our three Nigerian annotators belong to the Yoruba, Igbo, and Hausa ethnic groups; App. C.5).

Annotators rate each feature's prevalence using eWAVE's 1-4 scale, which we map to heuristic sampling probabilities: 4 = pervasive or obligatory (100% frequency), 3 = neither pervasive nor extremely rare (60%), 2 = exists but extremely rare (30%), and 1 = attested absence (0%).[2]

Maintaining an ordinal distinction is essential, as the SAE and dialect-specific features are not inherently mutually exclusive. For example, our British annotator gave "call"→"ring" a 3, because both "I'll call you" and "I'll ring you" are natural in British English. Our scale captures this nuance.

For morphosyntax, our pilot experiments revealed that eWAVE does not reflect modern usage patterns and lacks some country-level dialects (e.g., no Canadian English or no Standard British but "Southwest England", "East Anglian", etc.). As a result, we have our linguists re-annotate each eWAVE feature for all target dialects. For dialects present in eWAVE, we exclude features marked as absent in the original eWAVE database to reduce workload.

Finally, we have linguists annotate whether the model should follow each feature when generating text. For instance, the discourse marker "like" (e.g., I am, *like*, really tired.) was universally annotated as prevalent but not mandatory (3/4). At the same time, every annotator also indicated that models *should not* adapt this feature. This novel annotation dimension uniquely distinguishes DialectLLM from previous approaches.

Empirically, we see massive differences between eWAVE, user-appropriate ($Features_{\text{User}}$), and model-appropriate ($Features_{\text{Model}}$) morphosyntactic transformations; with per-feature alignment rate (agreement between eWAVE and annotator ratings) of only 34% and 16%, respectively. Moreover, our annotators indicate that models *should not produce* up to 90% of eWAVE features (Fig. 2). We also validate this distinction at the response level in Sec. 5.2. These results clearly highlight a large gap in prior work, which has taken eWAVE annotations at face-value. Our work provides the first effort to disentangle the morphosyntactic features of a spoken dialect from how LLMs should generate text.

3.2 Initial Dialog Generation

We generate parallel dialect-accurate conversations by transforming source SAE text into other dialects with DialectLLM. To ensure the SAE dialects possess features which differ from non-SAE dialects, we synthetically generate baseline SAE dialogs using the seed words identified in Sec. 3.1 as linguistically differentiating terms. We prompt SOTA LLMs to simulate a conversation between a user

[2]We follow the heuristic probabilities of eWAVE [4, 23].

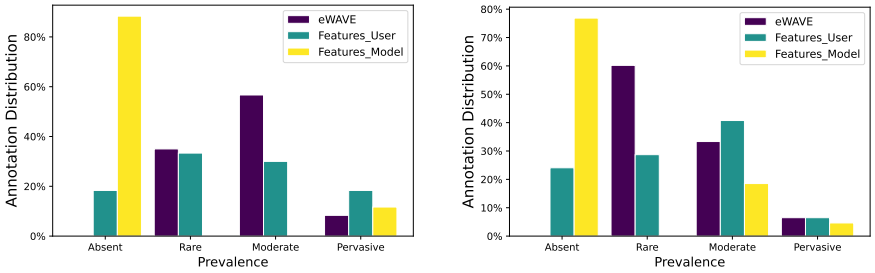


Figure 2: Prevalence rating distributions for AU (left) and PH (right) morphosyntactic features, comparing original eWAVE ratings with our newly annotated user- and model-appropriate ratings.

and an LLM agent based on the seed word. In **natural** conversations, the seed appears organically in the user's request (e.g., *"Will I need an umbrella today?")*, whereas in **indirect** conversations the user's initial request does not contain the seed and coaxes the model to generate it instead (e.g., "*What should I use to protect myself from the rain?")*.

3.3 Dialog Dialect Transformation

DialectLLM creates three datasets under different transformation settings: DialectLLM-OrthoLex (DialectLLM-OL), DialectLLM-User, and DialectLLM-Model. For each baseline dialog and dialect, d, we sample the set of valid transformations using weights assigned to the 1-4 scale. For example, "call"→"ring" was given a 3 in British English, corresponding to 60% frequency. For each dialog, there is a 60% chance this rule would be sampled. Note that most dialogs would not have this transformation applied either way, given the odds any single dialog contains "call" is quite low.

We found LLMs effectively apply lexical and orthographic rule-based transformations, both consistently applying the transformation rules while preserving semantic meaning and standard SAE grammar. We first transform the SAE dialog into the target dialect without explicit guidelines to fully utilize the LLM's parametric knowledge, then apply the revision process described in Sec. 3.4. We conduct these steps to create DialectLLM-OL.

In contrast, we found that even SoTA LLMs demonstrate a strong tendency to default to grammatical patterns that are accepted in SAE even when explicitly prompted otherwise. Thus, rather than providing all morphosyntactic rules at once, we follow prior work and sequentially apply each morphosyntactic rule one-by-one [4, 12]. This stepwise procedure helps ensure that each transformation is faithfully applied. Importantly, we have two separate morphosyntactic transformation guidelines, one for users ($features_{\text{User}}$) and a second for model ($features_{\text{Model}}$) generation. Thus, we conduct this process twice to create DialectLLM-User and DialectLLM-Model for their respective guidelines.

3.4 Quality Control & Revision

To ensure high-fidelity transformation, we engaged in iterative refinement in response to various challenges, which we discuss below.

eWAVE Guideline Regeneration. After providing example dialogs for DialectLLM-User & DialectLLM-Model to native linguists, many felt that the grammatical transformations were still too extreme or incorrectly applied. We traced these errors to the limitations of eWAVE. The database does not tailor specific features to individual dialects and as a result fails to provide adequately detailed transformation guidelines. For example, the eWave rule for "second person plural pronoun other than you" allows many different transformations (y'all, youse, all of you, you guys, etc.) while only one or two options may be legitimate. This deficit creates awkward or unnatural text: y'all is used where youse is correct or vice versa.

Plus, we observe that prior academic work [12] that attempts to generate transformation guidelines/examples with LLMs introduced errors, misdirecting models to produce non-natural or inauthentic dialectal outputs. For instance, for an eWAVE rule, "Leveling of the difference between present perfect and simple past: present perfect for StE simple past", the provided example is "She visited the museum yesterday" → "She has visited the museum.", where "yesterday" is unintentionally dropped. When an LLM is given this guide during the rule application step, it incorrectly learns to drop temporal expressions entirely rather than understanding that the rule concerns tense leveling. As a result, the model would erroneously drop any temporal expressions leading to unintentional

misguidance (e.g., "It rained on Monday." $\rightarrow$ "It has rained."). To correct these salient issues, we regenerate the guidelines and examples with iterative interactions and refinement processes with native linguists. This procedure is essential, considering that LLMs are very sensitive to the given examples and guidelines [25].

Orthographic & Lexical Refinement. Another critical issue is that our annotations are inexhaustive, meaning they do not cover all possible transformations—many valid transformations are not represented in the original Knowledge Base. We rely on LLMs to fill in these gaps parametrically, but LLMs occasionally have wrong knowledge about the lexical and orthographic differences between dialects, even in high-resource dialects. For instance, an LLM transforms *"eggplant"* to *"aubergine"* for AU dialect, probably due to its influence from GB, while *"eggplant"* is accepted in AU. Similarly, LLMs erroneously transform *"-ize"* conventions (e.g., organize) to *"-ise"* conventions (e.g., organise) for CA. We suspect that this occurs since CA follows GB spelling conventions with an exception of using *"-ize"*. Moreover, LLMs frequently apply GB spelling conventions for PH, despite PH following SAE orthography. We hypothesize that this error stems from lack of knowledge of low-resource dialects. We discuss our further revision steps to ensure accurate transformations in App. C.8.

4 DialectLLM-Bench

As a downstream application of DialectLLM, we introduce DialectLLM-Bench, a multi-dialectal benchmark to evaluate LLM capabilities on dialect identification and response completion tasks.

4.1 Classification: Dialect Identification

For LLMs to be effective in dialect-aware interactions, they must first correctly identify the user's dialect from the conversational context. Imagine that a shopping LLM agent is asked "Please purchase red pants size medium." If the user is British, they are asking the agent to purchase underwear while an SAE model would add trousers instead. If the model knew a-priori that the user is British, such a failure would be much less likely. Put simply, dialect identification failures can cascade to more severe failures. Though impossible to glean from this one sentence example, users naturally reveal their dialect over multi-turn conversations through their vocabulary, spelling, and grammar choices.

Can models discern the English dialect from natural language? Although an integral task, dedicated benchmarks for English dialect detection remain scarce [18]. This limitation stems from the scarcity of accurate, dialect-parallel datasets–a gap we address with DialectLLM. Through DialectLLM-Bench, we evaluate this capability by measuring identification accuracy across varying conversational turns, using $\{1,2,4,8\}$-turn dialogs from DialectLLM data. Models are given a dialog and prompted to choose the correct dialect from a set of options. We provide the prompt template in App. G.5.

4.2 Generation: Response Completion

Just as dialect identification is a necessary prerequisite to understanding the user's request, dialect-appropriate generation is essential to meet the needs of non-SAE speakers [15]. However, evaluating open-ended generation in this context is challenging. Traditional n-gram metrics like BLEU [26] are often poorly correlated with dialect appropriateness given the breadth of the valid output space; LLM -as-a-judge is unreliable, considering the lack of dialectal knowledge of LLMs (Details in App. D).

Hence, we develop a response completion task—given a conversation history and the user's current turn request, select the correct model response from N candidates, each representing a different dialect—and treat this task as a proxy for response generation capabilities. Note that we design the test set so that only one answer is valid in this task using the multi-label assignment mentioned in 4.3. For the prompt template, see App. G.6.

4.3 Multi-label Assignment

DialectLLM transforms an SAE dialect into a single target dialect. However, a conversation may be valid in multiple dialects, especially short ones where dialectal cues are limited. This creates a challenge for dialect identification and response completion tasks where multiple options may be valid choices. Addressing this challenge, we construct a $O \times L \times M \times D$ matrix (orthographic $\times$ lexical $\times$ morphosyntactic $\times$ dialect) for all valid transformations. When generating data, we track every transformation and validate each dialect against this set of transformations. We assign secondary labels to that dialog if *every* transformation is valid in additional dialects.

Table 2: Native-speaker preference evaluations. (a) DialectLLM win rate over Trans-EnV [12]. (b) Preference between DialectLLM-Model and DialectLLM-User for assistant (LLM) responses.

(a) DialectLLM **vs.** Trans-EnV

Eval. target	AU	IN	IR	NG	PH	SC
User utt.	0.99	0.99	0.97	0.98	1.00	0.99
Asst. resp.	1.00	1.00	0.99	0.96	1.00	0.98

(b) DialectLLM-Model (`M`) vs. DialectLLM-User (`U`)

Preferred	AU	IN	IR	NG	PH	SC
`M`	0.80	0.53	0.99	0.91	0.88	0.74
`U`	0.01	0.06	0.01	0.08	0.05	0.04

5 Experiments

5.1 Models and Dataset

We use 500 dialogs per turn (1, 2, 4, 8) and dataset (DialectLLM-{OL, User, Model}) resulting in approximately 54K test samples for evaluation (500 dialogs$\times$ 9 dialects$\times$ 3 datasets$\times$ 4 turn variants). We evaluate seventeen models, including Claude-{Haiku-4.5, Sonnet-4, Opus-4.1}, Qwen3-{0.6B, 1.7B, 4B, 14B, 8B, 32B, 235B-A22B} [27], Gemma3-{4B, 12B, 27B} [28], GPT-OSS-{20B, 120B} [29], and Deepseek-{V3.1, R1} [30, 31]. We also conduct supervised fine-tuning on Qwen3-8B and Gemma3-12B to see whether post-training enhances the dialectal capabilities of LLMs.

5.2 DialectLLM Quality Analysis

DialectLLM vs Prior Work. We evaluate DialectLLM's data quality against data generated by the most recent SoTA method, Trans-EnV [12], which provides a pipeline for synthetically generating non-SAE dialects. We construct single-turn SAE user-assistant dialogs and transform them into six non-SAE dialects shared by both frameworks. For each dialect, we recruit three native speakers, independent of the native linguists in Sec. 3, and ask which transformed dialog better represents (1) how a user would communicate and (2) how a model should respond in their English dialect. Preferences are aggregated via the median and we report the percentage of cases in which DialectLLM is preferred over Trans-EnV. DialectLLM substantially outperforms prior work, being preferred in approximately 99% of cases for both user and model utterances (Table 2(a)). Annotators consistently note awkward or incorrect grammatical transformations from Trans-EnV and compliment DialectLLM for attending to lexical and orthographic differences. See App. C.9 for full details.

Validating User–Model Split. We repeat this annotation procedure by comparing DialectLLM-User (`U`) and DialectLLM-Model (`M`) responses, this time asking annotators to focus exclusively on which response they prefer as a model response. Across six dialects, annotators prefer `M` in 80.65% of pairs on average, while `U` is preferred in only 4.23% (Table 2 (b)). The speakers comment that `U` responses apply informal or exaggerated features (e.g., "ye/youse", filler word such as "like", zero-article usage), that users may produce but assistants should avoid, hence greatly prefer `M` responses. Demonstrating remarkable consistency, features flagged by annotators in `U` responses as problematic were precisely the features identified by our native linguists as inappropriate for model-generated responses when building the transformation pipeline. See App. C.10 for full details.

5.3 LLM Performance on DialectLLM-Bench

Turning our attention now to DialectLLM-Bench performance, we find that even the frontier models such as Claude-Opus-4.1 or Claude-Sonnet-4 (with reasoning) struggle with dialect identification and generation tasks (Fig. 3). This finding illuminates prior work on model failures when engaging with non-SAE dialects [1, 12, 20], by highlighting a clear failure mechanism. If a model cannot infer whether a user is speaking Canadian, Nigerian, or Philippine English, it cannot reliably understand the user's request or adapt its response. Across the board, models perform worse on DialectLLM-User where morphosyntactic transformations are most extreme. Interestingly for classification (Fig. 10), models perform better on DialectLLM-Model than DialectLLM-OL, indicating that constrained morphosyntactic differences add information. Paradoxically, performance for small models *decreases* for the classification task as context (turn count) increases. We explore the relationships between turn count, model size, and performance below and conduct ablations to evaluate whether test-time compute for reasoning improves performance in App. E.6.

his phenomenon stems from two factors: (1) models possess relatively insufficient knowledge regarding the morphosyntax nuances across dialects and (2) orthographic and lexical variations are frequently shared across multiple dialects. This overlap results in more multi-labels for the $OrthoLex$

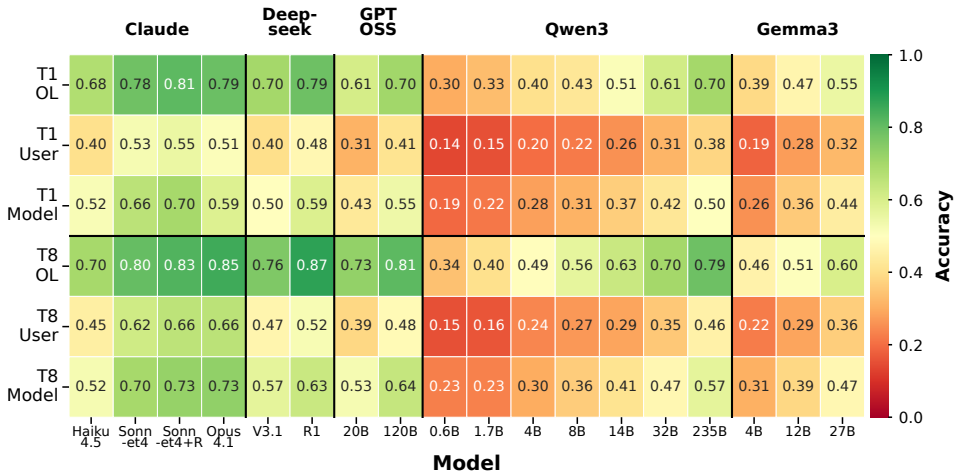


Figure 3: Model performance on DialectLLM-Bench generation task for turn 1 and 8 dialogs. We observe an overall trend of positive correlation between model size & accuracy and that models struggle when morphosyntactic features are added. "+R" indicates the model is equipped with reasoning. OL, User, and Model denote DialectLLM-OL, -User, and -Model dataset, respectively. Results across all turns for classification and generation are shown in Fig. 10 and Fig. 11, respectively.

dataset. The response bias of LLMs, shown in Sec. 5.5, are less penalized when multi-labels are prevalent, resulting in higher performance when morphosyntactic features are less applied.

5.4 Scaling Effects: Model Size vs Performance

We observe a clear positive linear correlation between model size and performance. We use estimated sizes for Claude-family models, which lack public parameter disclosures (Haiku: 20B; Sonnet: 175B; Opus: 2T)[3]. Fig. 12 shows that the performance of all models scale linearly with model size until plateauing at the largest scales. Though intuitive, this finding is noteworthy as LLMs develop dialect classification and generation capabilities despite being trained on unlabeled text without explicit dialectal metadata.

5.5 Context vs Performance

We expect the performance of LLMs to improve with longer context (more turns), as more context provide clearer dialectal signals. Fig. 4(a) shows a counter-intuitive pattern where smaller models perform better on short conversations. Investigating, we find that shorter conversations are often valid in multiple dialects (multi-label). Eight-turn dialogs, meanwhile, accumulate sufficient distinctive features and almost always has a single deterministic label.

Consequently, this ambiguity benefits models that make biased guesses. If a model defaults to high-resource dialects (e.g., SAE or British English), rather than genuinely analyzing dialectal features, it will score higher on ambiguous short conversations, where these dialects are often valid. All models smaller than 8B underperform the *GB-biased guess* (always selecting GB as the answer; App. E.1) for DialectLLM-User. Qwen3 models, in particular show large deficits on this task, with the 14B and 32B versions underperforming this baseline by 5% accuracy.

We explore these failure patterns as confusion matrices of predicted and actual labels for each model (App. E.3). Smaller models demonstrate a consistent bias towards high-resource dialects, regardless of the true dialect, and their success on short conversations is an artifact of label ambiguity, not genuine capability. This bias persists in even the largest models and is particularly pronounced for Canadian English. When averaging performance across turns, Claude-Opus-4.1, for example, correctly classifies Canadian English just 33% of the time, mislabeling 31% of dialogs as US (Fig. 14). This pattern holds for other large models—Deepseek-R1 (37% correct; 27% US), GPT-OSS-120B (31% correct, 27% US)—and is notably poor for Qwen3-235B (15% correct, 69% US) (Fig. 15).

[3] https://claude.ai/public/artifacts/0ecdfb83-807b-4481-8456-8605d48a356c

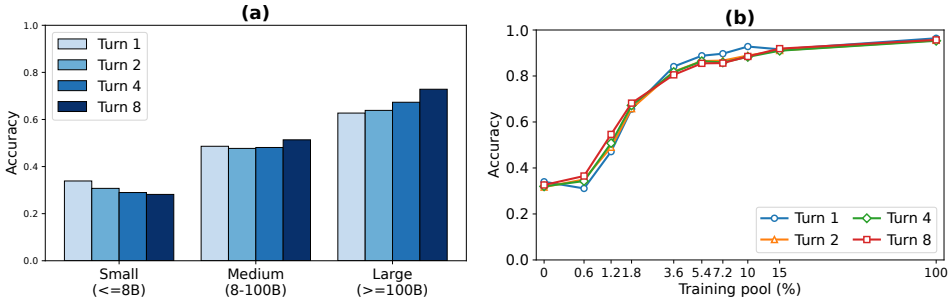


Figure 4: (a) Model performance by scale and context length averaged over all models and datasets for the classification task. (b) Qwen3-8B performance after fine-tuning on increasing fractions of the training pool for the generation task. The x-axis is displayed using $\log(1+x)$ spacing for readability.

5.6 Post Training with DialectLLM Data

Can the dialogs from DialectLLM be used to improve LLM's dialectal capabilities? We conduct supervised fine-tuning (SFT) on Qwen3-8B and Gemma3-12B for both tasks, using 2,500 dialogs per turn and dataset, resulting in approximately 365K training samples after filtration. We evaluate the fine-tuned models on DialectLLM-Bench. To minimize data leakage, we construct seed-disjoint splits, ensuring that no seed word appears in both the training data and the test sets.

Fine-tuning substantially improves model performance (App. E.4), with turn 1 accuracy boosting from 30% to approximately 90%, surpassing frontier models. To verify that these gains do not simply reflect data memorization, we train Qwen3-8B on progressively smaller subsets of the training pool. Performance improves gradually as more training instances are added, where the turn 1 accuracy for the response completion rises from 47% with 1.2% of the training pool to 93% with 10% (Fig. 4(b)). This trend suggests that DialectLLM data can help LLMs to learn comprehensive dialectal patterns.

5.7 Cross-Task Transfer

Does fine-tuning on classification improve generation or vice versa? To observe potential transfer learning, we post-train models on each task separately and report the results. Models trained for response completion (generation) demonstrate improved classification performance (Table 13). For example, Qwen3-8B-RC-0.15 performance on turn 1 classification improves by 10.9, 8.22, and 17.0 percentage points (pp) for DialectLLM-User, -Model, and -OL, respectively.[4] In contrast, we see a 10.6 pp decrease on average in response completion accuracy after classification SFT (Table 14).

These findings suggest dialect identification capabilities naturally emerge as models learns the distinct features of different dialects when generating responses (Table 13). Response completion requires fine-grained comparison between the candidates, building transferable discriminative capabilities. Classification, in contrast, involves assigning a single label to an entire dialog, a coarser task that may incur shortcut learning (e.g., detecting surface-level lexical cues) without developing a nuanced comparative reasoning required for response completion.

6 Conclusion

We introduce DialectLLM, a framework for generating parallel multi-dialectal English dialogs. Our primary theoretical contribution is to compartmentalize the dialect features present in user utterances, which models should understand, and those the model should produce in textual responses. We validate this bifurcation with expert native linguists, many of whom specialize in dialect-appropriate model responses, and find that as many as 90% of features in the eWAVE database *should not* be reproduced by models. The absence of this distinction in prior work risks perpetuating stereotypical and exaggerated dialect representations. Whereas prior work targets a single dialect dimension such as vocabulary [22] or grammar [7, 20, 24], to our knowledge DialectLLM is the first generator to encompass all three dimensions of written dialects: lexical, orthographic, and morphosyntactic dimensions. The result is the most authentic and natural dialect representations to date, with annotators preferring DialectLLM over most recent prior work [12] around 99% of the time. Using these data, we construct DialectLLM-Bench to evaluate dialect identification and response completion. Surprisingly,

[4] We mainly report the numbers for models trained on 15% of the training pool to avoid overfitting. Full results in App. E.5.

even frontier LLMs struggle with both tasks, achieving below 70% accuracy across different turns and exhibiting strong bias toward high-resource dialects (US and GB). Finally, we show that DialectLLM data also serve as a scalable post-training resource, boosting dialectal-task performance.

Limitations & Future Work. DialectLLM focuses on transcribable, country-level English dialects. We leave natural extensions, such as speech/audio-linked variation including accents and phonetics, finer-grained within-country variations, and non-English language dialects to future work.

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A Broader Impact

DialectLLM provides a foundation for systematic study of LLM dialect capabilities, highlighting both the scale of current deficits and a path toward improvement. As conversational AI becomes ubiquitous, ensuring equitable performance across the world's English dialects is not merely a technical challenge but an ethical imperative. The over one billion non-SAE English speakers deserve AI systems that understand and respect their linguistic identities. DialectLLM offers an important component for any future research seeking to generate, benchmark, or train LLMs to support the multitude of English dialects.

B Human Annotation Instructions

B.1 Lexical and Orthographic Annotations

Instruction Template (Irish English)

Task: Compare the original (US English) word/spelling with the transformed (Irish English) variant on a 1-4 scale:

1. **Feature is not used.** The transformed word/spelling is never correct in Irish English.
2. **Feature exists but is rare.** The word/spelling might be understood but is outdated, region-specific, or influenced by other dialects. An LLM/voice agent should understand this term but not generate it.
3. **Feature exists and is commonly used in colloquial speech.** The word/spelling is widespread in informal Irish English but may coexist with other variants. Both forms are acceptable.
4. **Feature is required in this locale.** Using the US variant would sound unnatural or incorrect. The Irish variant is the standard and expected form.

Examples:

US	IE	note	Rating	Explanation (Optional)
Excellent, Great	Savage	lexical		
Fruitcake	Barmbrack	lexical		
-or (e.g., color, favor)	-our (e.g., colour, favour)	orthographic		

Figure 5: Guidelines provided to Irish annotators for evaluating lexical and orthographic variations. Examples are truncated for brevity.

B.2 Morphosyntactic Feature Annotations

Instruction Template (Nigerian English)

We are working with eWAVE, a database of English dialect grammars, to build models to generate Nigerian-sounding phrases. This is important for two reasons:

1. **LLMs/conversational voice agents Understands non-US dialects:** Work with lexical variation found that LLMs/voice agents frequently misunderstood non-US terminology. When a user speaks with LLMs/voice agents, do they understand what the user is asking? Importantly, LLMs/voice agents should understand a variety of ways of speaking that are not necessarily "proper." In the US, users might say "ain't" in place of "is not," or use "like" as a filler word when speaking. The model should understand that this word is purely filler and the user is not making a simile.

2. **LLMs/voice agents respond with dialect-appropriate grammar:** Similarly, it is important that the models respond in a grammatically correct manner.

Task
Using eWAVE's list of grammar rules, we have created two versions of a sentence. The original sentence represents how the grammar rule in standard US English. The transformed sentence provides the sentence using this grammatical transformation. **Some example phrases are specific to certain regions. Please review the rules and explain any dialect-specific phrases that apply to them.** Please rate each rule by comparing the original and the transformed sentences on a 1-4 scale:

1. **Feature is not used.** The transformed sentence is never correct in this locale.
2. **Feature exists but is rare.** It might be characteristic of a specific region or an antiquated way of speaking that is outdated. An LLM/voice agent should understand this phrasing.
   US example: Double negatives: "He ain't never said that"
3. **Feature exists and is commonly used in colloquial speech.** However, it is not required or even standard. It might be characteristic to a specific region of the country.
   US example: Contractions: "ain't" in place of "is not" or "cuz" in place of "because"
4. **Feature is required in this locale.** It would sound unnatural if the speaker used the original sentence instead of the transformed sentence.
   US example: Possessive: The transformed sentence uses to "my object" when the original sentence uses "me object"

[Tips for Model Mirror? Column]
For each feature listed, please evaluate LLM/voice agent's capabilities in the specified dialect and fill in the columns as follows:

Model Mirror? Column: Write "Y" (Yes) or "N" (No) to indicate whether it would be better for the model to generate or respond with text or speech output including the feature for Nigerian English dialects.

Transformation Rule	Original Sentence	Transformed Sentence	Description	Rating (1-4)	Model Mirror?
Me instead of I in coordinate subjects	Sarah and I are going to the market.	Sarah and me are going to the market.	This transformation rule replaces the subjective pronoun "I" with the objective pronoun "me"...		

Figure 6: Guidelines provided to Nigerian annotators for evaluating morphosyntactic variations. Examples are truncated for brevity.

C Details on Data Generation

C.1 Dialectal Transformation Features

Table 3: *Dialectal transformation types in DialectLLM.* The table illustrates lexical, orthographic, and morphosyntactic variations that distinguish dialects.

Transformation	Description	Examples
Lexical	Lexical transformations refer to differences in vocabulary or the specific words used to refer to the same concept.	- Soda / Pop - Pants / Trousers - Umbrella / Brolly
Orthographic	Orthographic transformations involve variations in the spelling of words. The pronunciation and meaning are generally the same, but the written form differs.	- Color / Colour - Theater / Theatre - Realize / Realise
Morphosyntactic	Morphosyntactic transformations are about differences in grammar, including how words are formed and how they are arranged in sentences.	- Y'all / Youse / You guys - I like me shoes. (I like my shoes in SAE)

Table 4: Examples of dialectal feature transformations across lexical, orthographic, and morphosyntactic features. Text in red indicates specific modifications of the corresponding feature.

	Example Sentence
Original (SAE)	I realized the color of the theater poster was different from the sign, so we plan to get takeout at the gas station. Are you coming? (assuming the user is asking to two or more people)
+ Lexical	I realized the color of the theater poster was different from the sign, so we plan to get **takeaway** at the **servo**. Are you coming?
+ Orthographic	I **realised** the **colour** of the **theatre** poster was different from the sign, so we plan to get **takeaway** at the **servo**. Are you coming?
+ Morphosyntactic	I **realised** the **colour** of the **theatre** poster was different **to** the sign, so we plan to get **takeaway** at the **servo**. Are **youse** coming?

A visual depiction of the different transformations that occur in DialectLLM is shown in Table. 4. Lexical transformations involve the substitution of vocabulary, such as replacing "takeout" to "takeaway" and "gas station" to "servo". Orthographic transformations modify spelling conventions, such as converting "color" to "colour" and "realized" to "realised". Finally, morphosyntactic transformations address grammatical variations, including prepositional changes (e.g., "different from" to "different to") and the adoption of dialectal pronouns like "youse".

C.2 Examples of eWAVE Features

Table 5: Exemplar eWAVE features. The corresponding examples and guidelines generated in DialectLLM with LLMs along with iterative refinement partnering with linguists.

Rule	Original	Transformed	Description
She/her used for inanimate referents	The car is beautiful, and it runs perfectly.	The car is beautiful, and she runs perfectly.	This transformation rule involves replacing neuter pronouns (it/its) with feminine pronouns (she/her/hers) when referring to inanimate objects. Instead of using the standard ”it” to refer to non-living things like vehicles, machines, or objects, the transformed text personifies these items by assigning them feminine gender through pronoun usage. This creates a more personalized or affectionate tone toward inanimate objects, similar to how ships are traditionally referred to as ”she” in maritime contexts.
Me instead of I in coordinate subjects	Sarah and I are going to the market.	Sarah and me are going to the market.	This transformation rule replaces the subjective pronoun ”I” with the objective pronoun ”me” in coordinate subject constructions. When ”I” appears as part of a compound subject joined by ”and” (such as ”Sarah and I”), it is changed to ”me” (becoming ”Sarah and me”), even though the pronoun is functioning as the subject of the sentence. This represents a departure from standard English grammar, which requires the subjective case ”I” for subjects, in favor of using the objective case ”me” in these coordinate constructions.
Like as a focussing device	He was very angry.	He was, like, very angry.	This transformation involves inserting the word ”like” as a discourse marker or hedge, typically surrounded by commas, to focus attention on or emphasize the phrase that follows. The word ”like” functions as a parenthetical interjection rather than in its traditional grammatical roles, serving to highlight or draw special attention to the subsequent description or statement. This usage is characteristic of informal spoken English and creates a slight pause that signals the importance or precision of what comes next.

C.3 Examples of Different Datasets

Table 6 illustrates the distinct characteristics of our three transformation approaches. DialectLLM-OL captures lexical substitutions (parking lot $\rightarrow$ car park, gas station $\rightarrow$ service station) and orthographic aspects (kilometers $\rightarrow$ kilometres) consistent with Australian English. DialectLLM-User incorporates features including plural forms of you (youse) and discourse markers like “like”, resulting in heavily marked text that reflects the full range of attested dialectal features, while human annotators flagged that these features should not be mirrored by LLMs. DialectLLM-Model takes the middle ground by applying a subset of morphosyntactic features based on human annotation, which highlights our framework’s distinction between features that should be understood versus generated.

Table 6: Comparison of example transformation from prior works (Multi-Value [4] and Trans-EnV [12]) and DialectLLM-OL, DialectLLM-User, DialectLLM-Model datasets. Bold text denotes dialectal modifications from SAE for DialectLLM data.

User Input	I'm planning a cross-country road trip next month and I'm trying to figure out the best way to handle meals along the way. I don't want to rely entirely on fast food chains, but I also don't want to pack everything myself. What are some good options for finding decent food when you're driving through small towns or rural areas?
SAE	That sounds like an exciting adventure! For road trips through small towns and rural areas, roadside restaurants are often your best bet for authentic, home-cooked meals. These local gems usually offer hearty portions at reasonable prices and give you a real taste of regional cuisine. Look for places with full parking lots - that's usually a good sign that locals eat there regularly. Many of these establishments are family-owned diners or cafes that have been serving travelers and community members for decades. You can also use apps like Yelp or Google Maps to find highly-rated local spots along your route, or ask for recommendations at gas stations when you stop to fuel up.
Multi-Value	That sound an exciting adventure! Road trips small towns and rural areas, roadside restaurants are often your most good bet authentic, home-cooked meals. These local gems usually offers hearty portions reasonable prices and give you a real taste regional cuisine. Look places full parking lots - that's usually a good sign that eats there regularly. Many these establishments are family-owned diners or cafes has been serving travelers and community members decades. You can also use apps Yelp or Google Maps to find highly-rated local spots your route, or ask recommendations gas stations when you stops to fuel.
Trans-EnV	That is sounding like a, like, exciting adventure, but! For road trips through small towns and rural areas, roadside restaurants, which usual offer, like, hearty portions at reasonable prices and give you a real taste of regional cuisines, are often your, like, goodest bet for authentic, home-cooked meals, but. Look for places with full parking lots, and avoid places with empty lots - those mustn't be good since locals clear don't eat there regular. There's many family-owned diner or cafes that served travelers and community members for decades. You can also use apps like Yelp or Google Maps to find, like, high-rated local spots along your route, or ask for recommendations at gas stations when you stop to fuel up.
DialectLLM-OL	That sounds like an exciting adventure! For road trips through small towns and rural areas, roadside restaurants are often your best bet for authentic, home-cooked meals. These local gems usually offer hearty portions at reasonable prices and give you a real taste of regional cuisine. Look for places with **chockers car parks** - that's usually a good sign that locals eat there regularly. Many of these establishments are family-owned diners or cafes that have been serving **travellers** and community members for decades. You can also use apps like Yelp or Google Maps to find highly-rated local spots along your route, or ask for recommendations at **service stations** when you stop to fuel up.
DialectLLM-User	That **there** sounds like an exciting adventure! For road trips through small towns and rural areas, roadside restaurants are often your most best bet for **, like,** authentic, home-cooked meals. These local gems here usually offer hearty portions at reasonable prices and give youse a real taste of regional cuisines. Look for places with **chockers car parks** - that's usually a good sign that locals eat there regularly. Many of these establishments **here** are**, like,** family-owned diners or cafes that have been serving **travellers** and community members for decades. **Youse** can also use apps like Yelp or Google Maps to find more highly-rated local spots along your route, or ask for recommendations at **service stations** when **youse** stop to fuel up.
DialectLLM-Model	That **there** sounds like an exciting adventure! For road trips through small towns and rural areas, roadside restaurants are often your best bet for authentic, home-cooked meals. These local gems here usually offer hearty portions at reasonable prices and give you a real taste of regional cuisine. Look for places with **chockers car parks** - that's usually a good sign that locals eat there regularly. Many of these establishments **here** are family-owned diners or cafes that have been serving **travellers** and community members for decades. You can also use apps like Yelp or Google Maps to find highly-rated local spots along your route, or ask for recommendations at **service stations** when you stop to fuel up.

C.4 DialectLLM Algorithm

Algorithm 1 DialectLLM Pipeline

Input: Source Dialect S (SAE), Target Dialects $\mathcal{T}$, eWAVE Rules $\mathcal{R}$
Output: Multi-dialectal Dataset $\mathcal{D}_{final}$
Initialize: $\mathcal{D}_{final} \leftarrow \emptyset$
// Step 1: Wordbank Construction (Lexical/Orthographic)
for all $t \in \mathcal{T}$ **do**
  $\mathcal{C}_t \leftarrow \text{LLM}_{gen}(\text{RetrieveDiff}(S, t))$
  $\mathcal{W}_t \leftarrow \{(w_s, w_t) \in \mathcal{C}_t \mid \text{LLM}_{ver}(w_s, w_t) > \tau_{ver}\}$
end for
// Step 2: Seed Dialog Generation
$\mathcal{W}_{seed} := \bigcup_{t \in \mathcal{T}} \mathcal{W}_t$
for all $(w_s, w_t) \in \mathcal{W}_{seed}$ **do**
  $d_{nat} \leftarrow \text{GenDialog}(w_s, \text{mode} = \text{Natural})$
  $d_{ind} \leftarrow \text{GenDialog}(w_s, \text{mode} = \text{Indirect})$
  $\mathcal{D}_{src} \leftarrow \mathcal{D}_{src} \cup \{d_{nat}, d_{ind}\}$
end for
// Step 3 & 4: Guideline Refinement & Human Annotation
$\mathcal{G} \leftarrow \text{RefineGuideline}(\mathcal{R})$
$\mathcal{A}_{OL} \leftarrow \text{HumanAnnotation}(\mathcal{W}_t)$
$\mathcal{A}_{Morph} \leftarrow \text{HumanAnnotation}(\mathcal{G})$
for all $d \in \mathcal{D}_{src}$ **and** $t \in \mathcal{T}$ **do**
  // Step 5: OrthoLex Application
  $d' \leftarrow \text{ApplyOrthoLex}(d, \mathcal{W}_t)$
  // Step 6: Morphosyntactic Transformation
  for all rule $r \in \mathcal{A}_{Morph}$ applicable to d' **do**
    $p_{rate} \leftarrow r$.prevalence {Scale 1-4}
    if DialectLLM-Model and r.Model Mirror $= False$ **then**
      $p_{rate} \leftarrow 0$
    end if
    $$\pi_{inject} \leftarrow \begin{cases} 1.0 & \text{if } p_{rate} = 4 \\ 0.6 & \text{if } p_{rate} = 3 \\ 0.3 & \text{if } p_{rate} = 2 \\ 0.0 & \text{otherwise} \end{cases}$$
    if Random() $< \pi_{inject}$ **then**
      $d' \leftarrow \text{ApplyRule}(d', r)$
    end if
  end for
  // Step 8: Quality Control & Revision
  $C_{changes} \leftarrow \text{ExtractTransformation}(d, d')$
  for all $c \in C_{changes}$ **do**
    if $\mathcal{A}_{OL}$.rating$(c) < 4$ and $\mathcal{A}_{OL}$.rating(c).source $=$ LLM **then**
      $\mathcal{A}_{OL}$.rating$(c) \leftarrow 1$
    end if
    $$p_{rev} \leftarrow \begin{cases} 0.0 & \text{if } \mathcal{A}_{OL}.\text{rating}(c) = 4 \\ 0.4 & \text{if } \mathcal{A}_{OL}.\text{rating}(c) = 3 \\ 0.7 & \text{if } \mathcal{A}_{OL}.\text{rating}(c) = 2 \\ 1.0 & \text{otherwise} \end{cases}$$
    if Random() $< p_{rev}$ **then**
      $d' \leftarrow \text{RevertChange}(d', c)$
    end if
  end for
  if $\exists c, c \notin C_{changes}$ and $c \in \mathcal{W}_t$ **then**
    ApplyRule(d', c) {Probabilistic Application }
  end if
  $\mathcal{D}_{fin} := \bigcup d'$
end for

C.5 Annotator Recruitment and Assignment.

For Irish (IR), Scottish (SC), Indian (IN), Philippine (PH), and Nigerian (NG) English dialects, 3 native speakers or linguists per locale were recruited, with majority voting employed to determine the final labels for each attribute. For Canadian (CA), Australian (AU), British (GB), and SAE (US), one native linguist, who specialize in dialect-appropriate LLM generation, per locale was assigned with *iterative feedback cycles* to ensure annotation quality and consistency. We put significant effort on linguist recruitment for quality and language/ethnicity-wise diversity. In more detail, we ensure the quality of annotations with several attention checks and recruit people from diverse backgrounds to capture intra-country dialect variation, for instance, speaking Cebuano, Ilocano, and Tagalog along with English for PH and ethnicity of Yoruba, Igbo, and Hausa for NG.

C.6 Visual Depiction of Wordbank Aggregation

Continuing from Sec. 3.4, we show a visual depiction of the aggregated wordbank matrix filled by LLMs and humans.

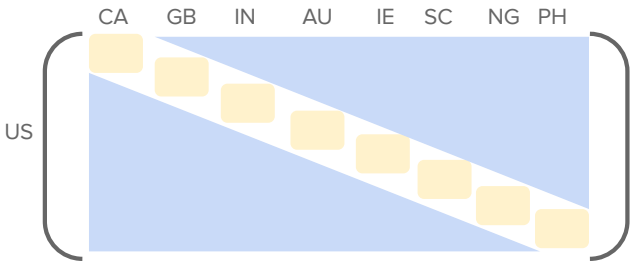


Figure 7: LLM-based wordbank matrix completion. Using human annotations as few-shot demonstrations (diagonal yellow), we prompt an LLM to fill out missing entries in the dialectal mapping matrix (blue regions).

C.7 Annotation Rating Alignment and Distribution (eWAVE vs DialectLLM)

Continuing from Sec. 3.1, we report the feature-level alignment rate (Table 7) and the general distribution (Fig. 8) of human ratings compared to the ratings present in the original eWAVE database. We observe a huge discrepancy in both alignment and distribution, where these differences become more pronounced for the $Features_{Model}$. Notably, the alignment gets as low as 18% and 2% alignment for $Features_{User}$ and $Features_{Model}$, respectively. This finding suggests that the mismatch can be the primary cause of the perceived unnaturalness in eWAVE-based generation.

Table 7: Alignment rates between eWAVE features and human annotations for $Feature_{User}$ and $Feature_{Model}$. Low alignment across all dialects underscore the inadequacy of using raw eWAVE ratings for natural dialect transformation.

Dialect	$Features_{User}$ (%)	$Features_{Model}$ (%)
Australian English	41.67	1.67
Indian English	18.00	2.00
Irish English	40.22	27.17
Nigerian English	41.18	31.37
Philippine English	33.33	7.41
Scottish English	27.27	25.45

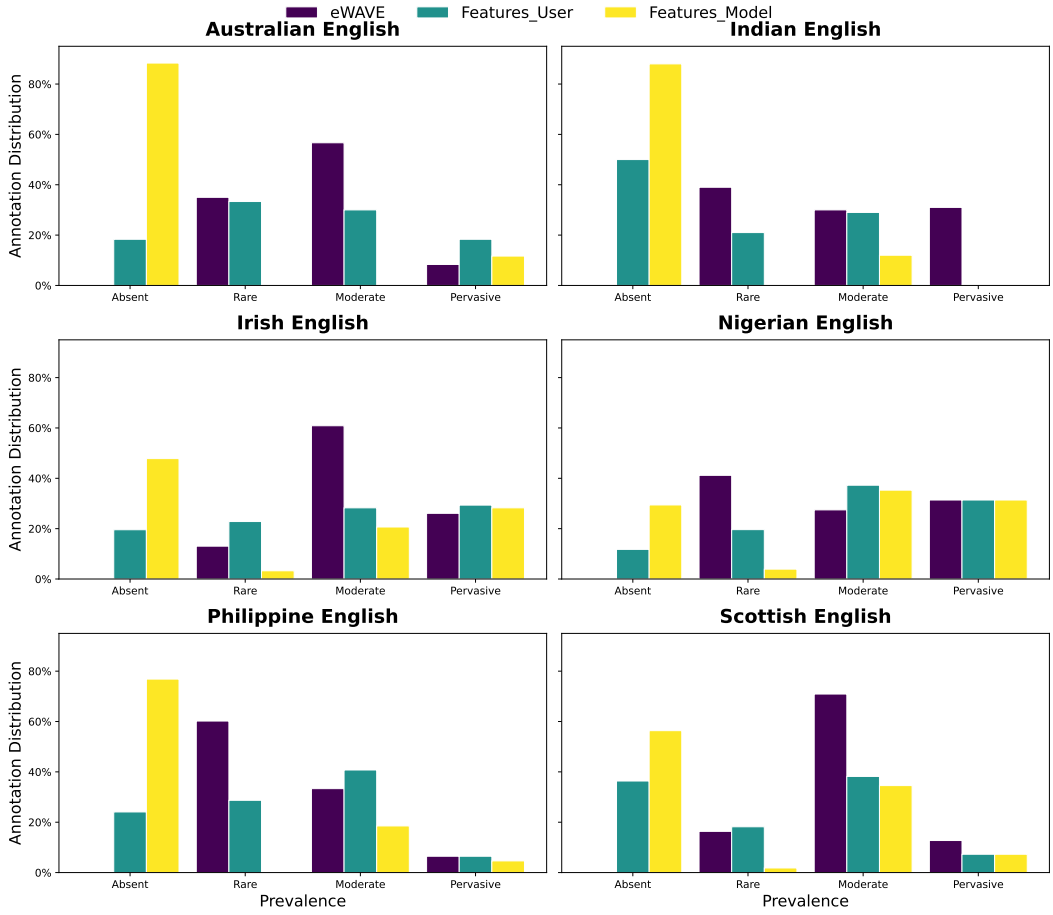


Figure 8: Annotation rating distribution of eWAVE, $Features_{User}$, and $Features_{Model}$ for dialects that exist in the eWAVE database. We observe clear discrepancies between the new ratings from native linguists and the original eWAVE annotations, with larger differences for $Features_{Model}$.

C.8 Additional Data Quality Assurance

Continuing from Sec. 3.4, first, we aggregate mappings present across all SAE $\leftrightarrow$ other dialects wordbanks. We provide an LLM human ratings from Sec. 3.1 as few-shot demonstrations and calibration reference and prompt to rate the missing part of the matrix (See Fig. 7 for visual depiction and App. G.4 for prompt templates). This matrix assists with comprehensive coverage of words. Second, to ensure that the initial transformations (fully by LLMs) are valid, an LLM, guided with human ratings as few-shot reference, is prompted to rate the transformations that have happened. Note that human ratings (R_h) are prioritized over LLM ratings (R_m) in all cases. For mappings that R_h don't exist, we apply a conservative threshold, where only those with $R_m = 4$ are accepted as valid transformations, while a rating of 1 to 3 results in reversion to the original SAE form.

If the initially executed transformation is rated 4 by humans or LLMs, the transformation is retained. If $R_h = 3, 2$, or 1, we revert the change probabilistically (40, 70, or 100% respectively) and $R_m \leq 3$, revert. For transformations that did not occur initially (false negatives with reference from wordbank, we apply them in a probabilistic manner ($R_h = 2, 3$, or $4 \rightarrow 30, 60$, or 100%, respectively; $R_m = 4 \rightarrow 100\%$). As a result, we create 3 final datasets, DialectLLM-OL, DialectLLM-User, and DialectLLM-Model, where each dataset has different magnitude of morphosyntactic transformations.

C.9 Comparison to Prior Work: Human Preference Annotations

We compare our data generation method with Trans-EnV [12], a leading non-SAE dialect generator. We apply their method on the synthetically generated conversations described in Section 3 using their open-source repository[5]. Off the shelf, we observed that their prompt struggled with the "USER: ... ASSISTANT:..." text format, so we modified the prompt slightly to include the following two lines:

> - CRITICAL: Your output must contain EXACTLY ONE complete conversation with ONE "USER:" and ONE "ASSISTANT:". Do NOT include the original conversation in your output.
> - CRITICAL: Your output must ALWAYS contain both USER: and ASSISTANT: sections, even if only one is transformed.

With this change, we observe strong adherence to the model's prompt and a faithful application of the eWave transformation rules through Trans-EnV.

We partner with a 3P data annotation service to recruit three annotators to compare 120 DialectLLM and Trans-EnV generated conversations. We present these two data transformations side-by-side and ask annotators to provide two preference annotations on 1 to 5 scale (one for the user utterance and a second for the model response):

1. V1 is **greatly preferred** over V2
2. V1 is **somewhat preferred** over V2
3. No clear preference between V1 and V2
4. V2 is **somewhat preferred** over V1
5. V2 is **greatly preferred** over V1

Order was randomly shuffled to ensure no annotation bias. We provide the following additional annotation instructions:

> **For the USER annotations:** Which version is more representative of how users from your country speak in day-to-day conversation? We are trying to capture colloquial expressions (e.g., "y'all" instead of "you all") and other features of the spoken dialect. These may include forms that are not technically "proper" but are commonly used and should be understood by the model.
>
> **For the ASSISTANT annotations:** Which model response is more appropriate for your country? While users may employ informal or nonstandard constructions when speaking (e.g., double negatives such as "ain't got none"), the model generally should not. Use your best judgment to decide what a dialect-appropriate model response should sound like.

[5] https://github.com/jiyounglee-0523/TransEnV

> 'Strong' indicates high confidence or a clear qualitative gap; 'moderate' indicates a noticeable but smaller difference. Please try to avoid selecting (3) unless there is truly no clear difference. If both transformations are bad, prefer the option that is less bad.

We aggregate these preference annotations by taking the median value between the three annotators and present the percentage of time DialectLLM is preferred over Trans-EnV in Table 2(a). Table 8 provides detailed results. Importantly, 99% of the time DialectLLM is the preferred generator for both the User and the Model. Frequently, our annotators highlighted the awkwardness of the grammatical transformations of Trans-EnV and complimented the proper lexical and orthographic usages of DialectLLM, e.g., "V2 (DialectLLM) is clear and natural, and uses the correct term "dickey" (common in Indian English). V1 (Trans-EnV) is grammatically broken and awkward, so it doesn't sound like normal speech."

Collectively, these findings emphasize the limitations of eWAVE as a source of dialect transformation guidelines. While both DialectLLM and Trans-EnV rely upon eWAVE as a ground truth source for morphosyntactic transformations, we worked with native linguists to heavily prune the set of valid transformations. As a result, DialectLLM imposes substantially fewer grammatical transformations, preserving the naturalness of the conversation. While any given transformation in eWAVE may be valid in the abstract, applying many rules simultaneously results in a grammatically fragmented and unnatural transformation.

Table 8: DialectLLM vs Trans-EnV Preference Distribution (%)

Role	Preference	AU	IN	IE	NG	PH	SC
User	DialectLLM greatly preferred to Trans-EnV	0.90	0.94	0.68	0.79	0.88	0.73
	DialectLLM somewhat preferred to Trans-EnV	0.09	0.05	0.28	0.19	0.12	0.26
	DialectLLM approximately equal to Trans-EnV	0.00	0.00	0.02	0.01	0.00	0.00
	Trans-EnV somewhat preferred to DialectLLM	0.01	0.01	0.01	0.00	0.00	0.01
	Trans-EnV greatly preferred to DialectLLM	0.00	0.00	0.00	0.01	0.00	0.00
Model	DialectLLM greatly preferred to Trans-EnV	0.93	0.99	0.69	0.81	0.99	0.62
	DialectLLM somewhat preferred to Trans-EnV	0.07	0.01	0.30	0.15	0.01	0.36
	DialectLLM approximately equal to Trans-EnV	0.00	0.00	0.01	0.02	0.00	0.02
	Trans-EnV somewhat preferred to DialectLLM	0.00	0.00	0.00	0.00	0.00	0.00
	Trans-EnV greatly preferred to DialectLLM	0.00	0.00	0.00	0.02	0.00	0.00

C.10 Validating User–Model Splits: Human Preference Annotations

Similar to Sec. C.9, we present native speakers with 100 side-by-side pairs, each containing an DialectLLM-User and DialectLLM-Model. We ask the annotators which response better represents how a model should respond in their dialect in a 5-point scale. The order of the options was shuffled to ensure no annotation bias. The quantitative results are shown in Table 9, where we observe a strong preference (80.65%) towards DialectLLM-Model for the model response, while the opposite direction holds for only 4.23% of the cases.

Table 9: Detailed DialectLLM-Model vs DialectLLM-User preference distribution. Native speakers compared 100 shuffled response pairs per dialect.

Preference	AU	IN	IE	NG	PH	SC
DialectLLM-Model greatly preferred to DialectLLM-User	0.35	0.15	0.68	0.68	0.42	0.09
DialectLLM-Model somewhat preferred to DialectLLM-User	0.45	0.38	0.31	0.23	0.46	0.65
DialectLLM-Model approximately equal to DialectLLM-User	0.19	0.41	0.00	0.01	0.07	0.22
DialectLLM-User somewhat preferred to DialectLLM-Model	0.01	0.06	0.01	0.04	0.05	0.04
DialectLLM-User greatly preferred to DialectLLM-Model	0.00	0.00	0.00	0.04	0.00	0.00

Beyond aggregate preferences, annotator comments consistently reveal that DialectLLM-User often preserves dialect features that native speakers perceive as natural for user utterances but inappropriate for assistant/model responses. Representative comments are shown in Table 10. These comments closely align with our feature-level annotations (Sec. 3.1), where features such as discourse-marker

like, colloquial second-person pronouns, zero-article usage, and pluralized uncountables are marked as user-understandable but not model-appropriate. Thus, the qualitative feedback helps explain the strong preference for DialectLLM-Model and provides a validation of the user–model split.

Table 10: Representative annotator comments for the DialectLLM-Model versus DialectLLM-User preference annotation. Display order was randomized during annotation; we replace references to the two variants for readability.

Dialect	Representative comments
IN	- DialectLLM-Model is greatly preferred because it avoids unnecessary filler words and maintains clear grammar suitable for an assistant response. - DialectLLM-Model is an excellent example of my dialect and is grammatically correct.
IR	- No issues with DialectLLM-Model, but DialectLLM-User is too colloquial. - DialectLLM-User contains examples that are not appropriate for written AI.
SC	- DialectLLM-Model is greatly preferred; DialectLLM-User is too informal for the assistant response. - DialectLLM-User uses grammar that the model should not use. For example, the use of *like*, *ye*, or *youse* is a behavior that the model should not copy.
NG	- DialectLLM-Model maintains a helpful and 'neighbourly' tone while correctly using the Nigerian spelling of *organise*. While DialectLLM-User attempts dialectal markers such as zero-article usage and pluralized uncountables, these result in an over-transformed or overly informal tone that the assistant should generally avoid.

D LLM Benchmarking for Open-Ended Dialectal Generation

Considering that the current LLMs are not well-equipped for dialectal capabilities, concrete evaluation of generation capabilities through LLMs in near impossible. Taking into consideration, we test the generation capabilities with a proxy task of response completion, where we give the candidate response to an LLM and prompt the model to choose the dialect-appropriate response as shown in Sec. 4.2. However, to get a glimpse of the current models' capabilities truly on generation when guided to output a certain dialect, we evaluate the output results of LLMs with different LLM-as-a-Judges (LLMaaJs) [19]. To do this, we for each target turn (e.g., turn 8), we provide the model with the preceding conversation history (turns 1 through 7) and the last turn's user utterance and prompt the model to respond in a specified dialect. For the generated output, we evaluate the models' performances with three LLMaaJs, using Claude-4-Sonnet), where we construct separate evaluators for each orthographic, lexical, and morphosyntactic features. Each judge is provided with human annotations as reference for their assessment and performs binary classification (valid or invalid), with an N/A option when the generated text contains no relevant markers.

LLMaaJ proves suboptimal for generation evaluation. This roots from two causes: (1) LLMs lack proper linguistic knowledge for dialects and (2) models exhibit high hallucination rates even on straightforward binary judgment [32]. Despite the limitation, we report LLMaaJ as a rough indicator for relative performance differences across different models. When computing the scores, we exclude the abstained (N/A) responses. This assessment introduces a slight advantage for smaller models, since when a model fails to produce coherent English entirely (e.g., outputting Tagalog when prompted for Philippine English or generating repetitive nonsense), which happens occasionally for smaller models, judges abstain rather than penalize. This likely explains the unexpected performance drop from Qwen3-14B to Qwen3-32B.

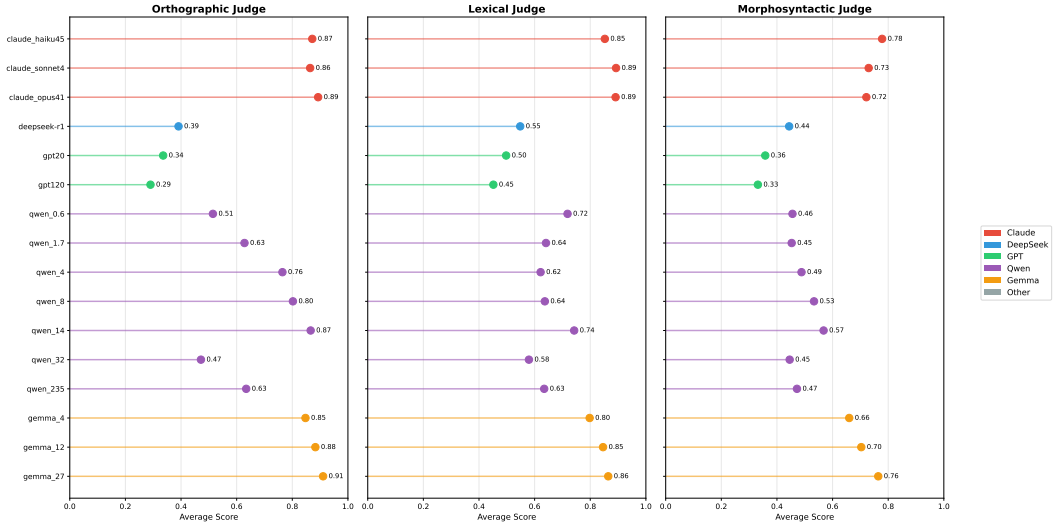


Figure 9: LLMaaJ evaluation results for open-ended generation. Scores across orthographic, lexical, and morphosyntactic judges confirm that performance scales with model size, consistent with other tasks.

As shown in Fig. 9, larger models within each model family generally outperform smaller variants, with Claude and Gemma demonstrating the strongest generation capabilities overall. Given LLMaaJ's limitations, these scores should be interpreted as relative comparisons, preferably within the model family, rather than absolute measures of dialect generation quality. Moreover, we can observe that LLMs are less capable of incorporating appropriate morphosyntactic features compared to lexical or orthographic features. We leave the development of generation metrics and dialectal post-training for better open-ended generation as important directions for future research.

E Additional Results

E.1 More Results from Sec. 5.3

We present the full results for the classification (identification) and generation (response completion) task in Fig. 10 and Fig. 11 across 1, 2, 4, and 8 turn dialogs. We observe similar trends for both classification and generation tasks, while models show a slight degradation in performance for the generation task compared to that of classification. For classification, to provide context for the results, we establish three baselines for the classification task across 1, 2, 4, and 8 turns. A *random guess* yields {.31, .27, .24, .22}, *SAE-biased guess* (always selecting SAE as the answer) yields {.21, .18, .17, .15}, and *GB-biased guess* yields {.37, .33, .29, .26} average accuracies, respectively.



Figure 10: Model performance on DialectLLM-Bench classification (Identification) task. We observe an overall trend of positive correlation between model size & accuracy and that models struggle when morphosyntactic features are added. OL, User, and Model denote the DialectLLM-OL, -User, and -Model dataset, respectively.

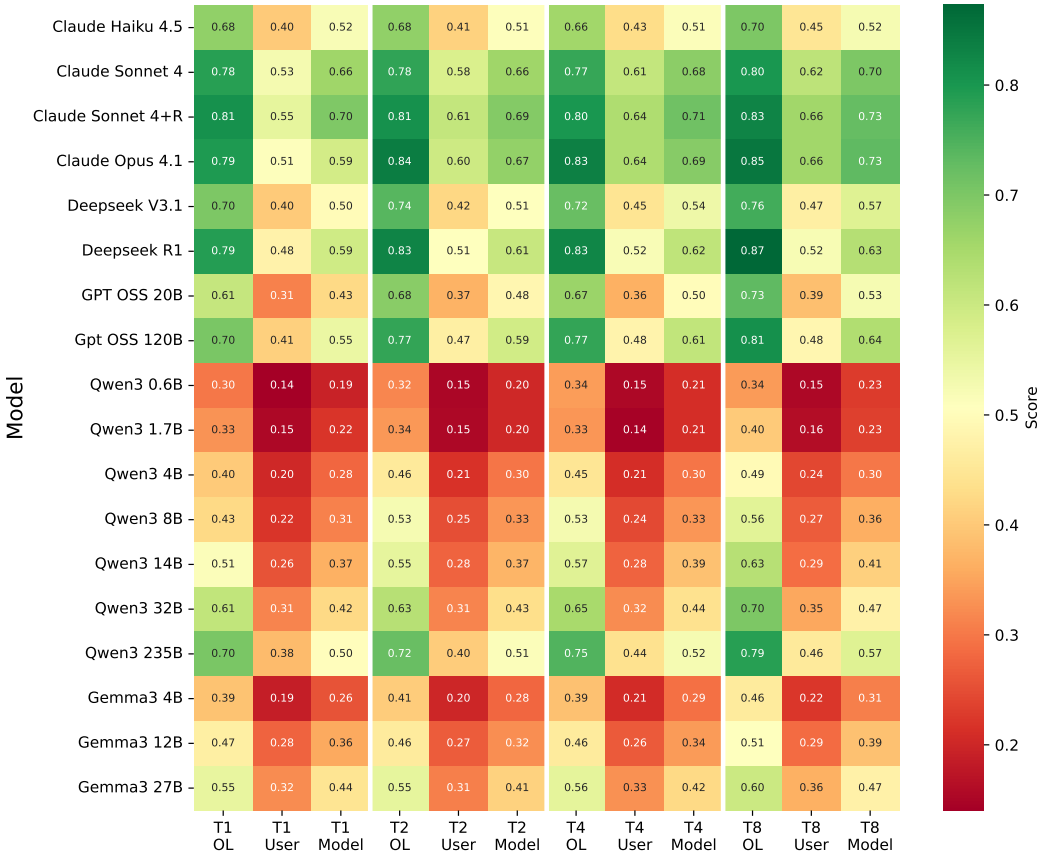


Figure 11: Model performance on DialectLLM-Bench generation (response completion) task. Similar trend to Fig. 10 is observed.

E.2 More Results from Sec. 5.4

We show the correlation between model size and performance for both classification and generation task on DialectLLM-Bench. We see a linear positive correlation within the model family and also in general across families.

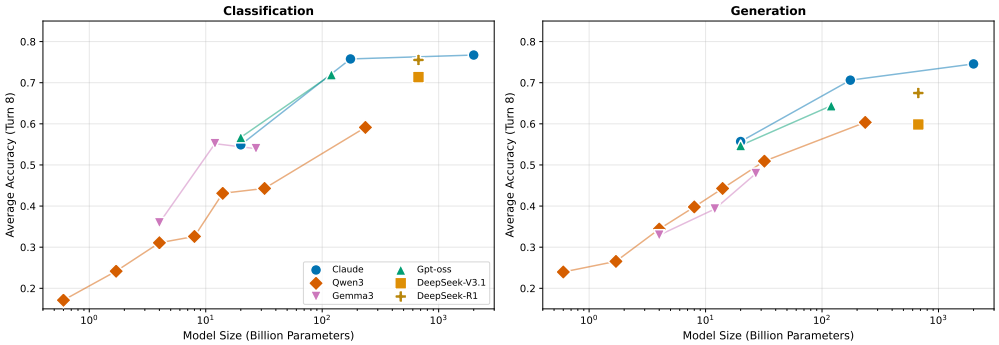


Figure 12: Scaling behavior of LLMs on dialectal classification and generation tasks. We show the model performance at turn 8, which has the most contextual cues and has the least multi-labels, to avoid giving credit for models outputting US or GB-biased answers. Accuracy scales linearly with model size.

E.3 More Results from Sec. 5.5

In this section, we show the confusion matrices on the classification task of different models. More context typically incorporates more dialectal cues, hence the likelihood of a unique gold label gets higher. Considering that a standard confusion matrix requires a single gold label, we report the confusion matrix for two representative models (Qwen3-1.7B and Claude-Opus-4.1) for 8-turn dialogs as a visual representation (Fig. 13). We also present the confusion matrices of different models aggregated across all turns and datasets in Fig. 14 and Fig. 15.

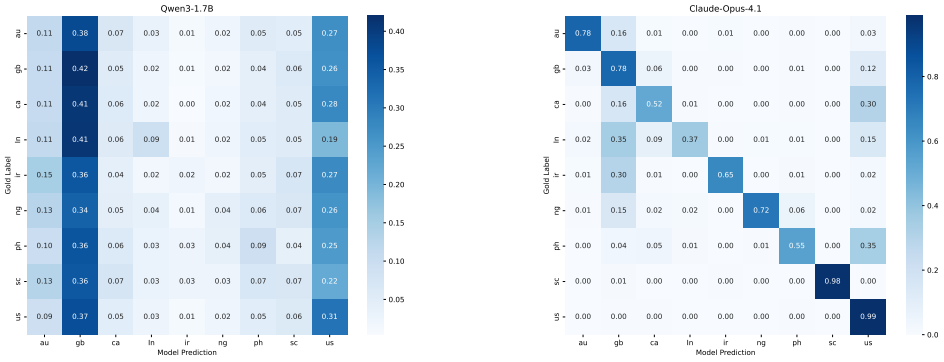


Figure 13: Confusion matrix for Qwen3-1.7B (left) and Claude-Opus-4.1 (right) on 8-turn dialogs; classification task. For Qwen3-1.7B, The vertical clustering in the US and GB columns suggests that the model is making default guesses to standard dialects rather than analyzing dialectal features. On the other hand, for Claude-Opus-4.1, the diagonal pattern indicates that the model is capable of genuinely identifying dialects.

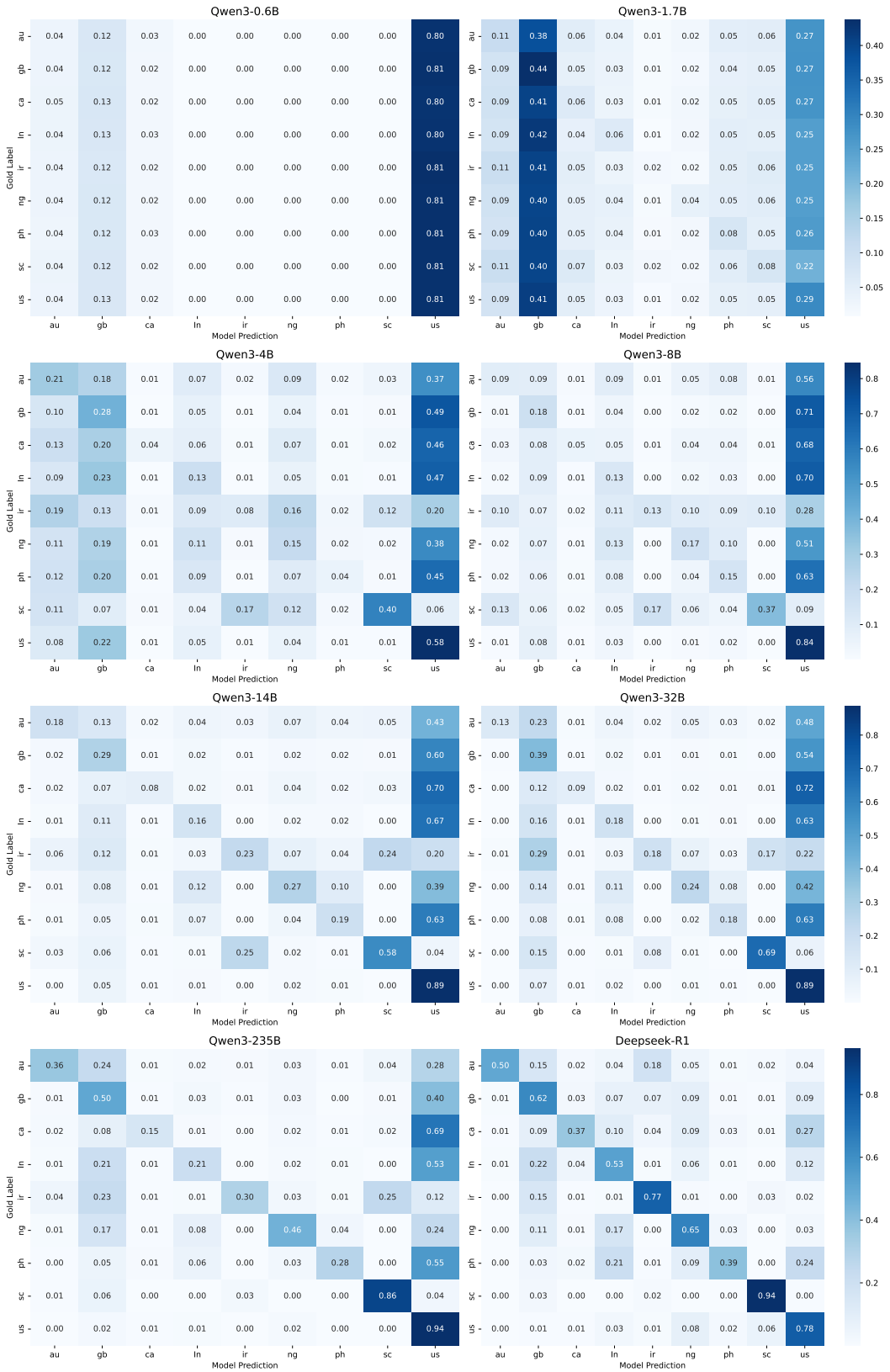


Figure 14: Confusion matrices of different models averaged over all turns (1, 2, 4, and 8); classification task. Smaller models show more bias towards high-resource, standard dialects (US, GB).

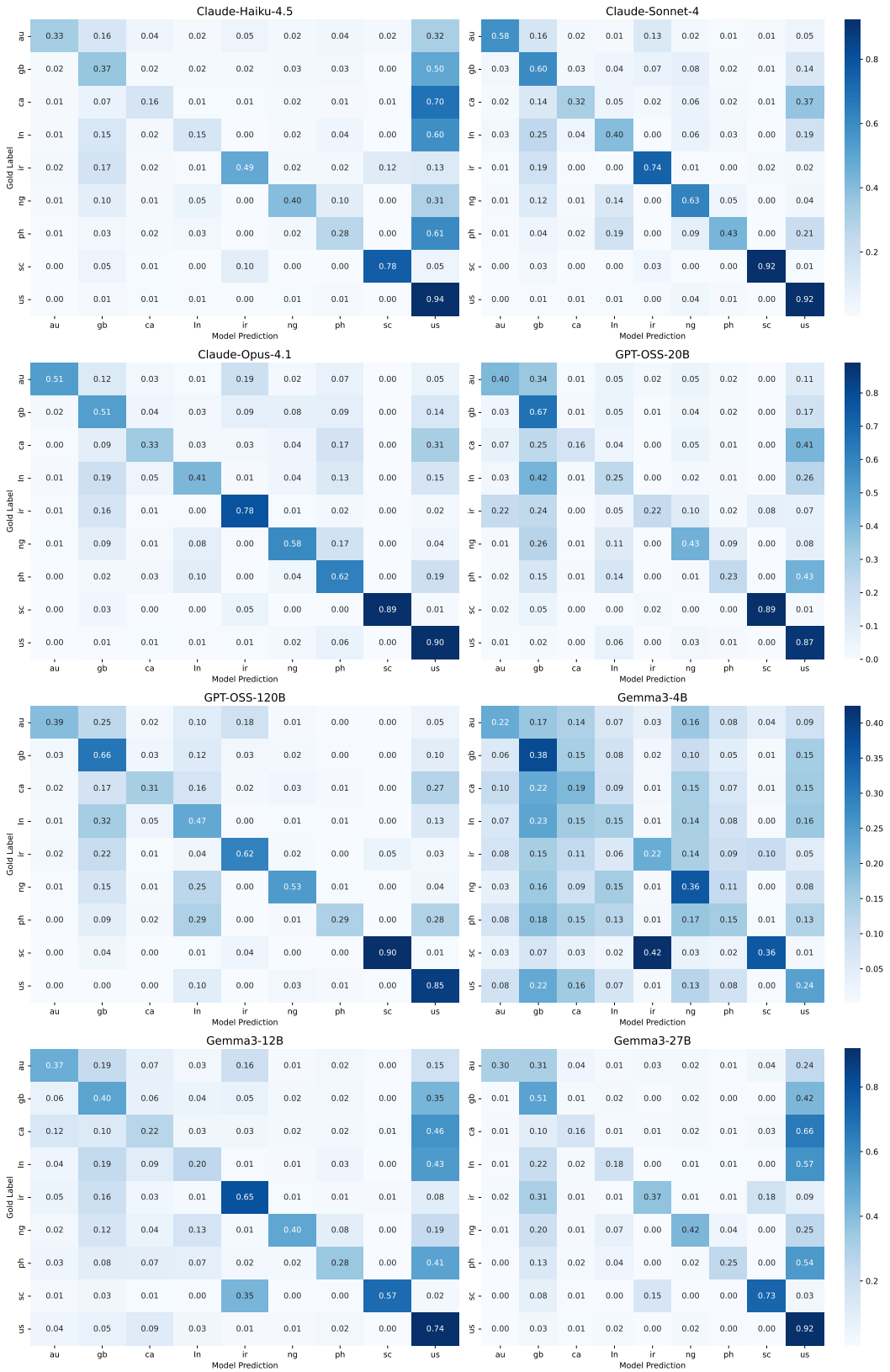


Figure 15: Confusion matrices of different models averaged over all turns (1, 2, 4, and 8); classification task. Smaller models show more bias towards high-resource, standard dialects (US, GB).

E.4 More Results from Sec. 5.6

Table 11: Model performance results on DialectLLM-Bench classification task after post-training. Models with suffix "-0.15" are models trained on 15% of the training pool and those with suffix "-full" are trained on the whole training data. OL, U, and M denote DialectLLM-OL, -User, and -Model, respectively.

	Turn 1			Turn 2			Turn 4			Turn 8		
Model	OL	U	M	OL	U	M	OL	U	M	OL	U	M
Qwen3-8B	0.353	0.237	0.429	0.326	0.240	0.385	0.318	0.265	0.377	0.319	0.287	0.373
Qwen3-8B-0.15	0.746	0.878	0.919	0.782	0.941	0.957	0.838	0.976	0.977	0.887	0.991	0.990
Qwen3-8B-Full	0.860	0.956	0.967	0.872	0.990	0.991	0.910	0.998	0.996	0.944	0.999	0.999
Gemma3-12B	0.535	0.410	0.622	0.511	0.436	0.605	0.507	0.450	0.599	0.546	0.493	0.617
Gemma3-12B-Full	0.875	0.967	0.967	0.882	0.990	0.990	0.915	0.998	0.997	0.949	1.000	0.999

Table 12: Model performance results on DialectLLM-Bench generation task after post-training. Models with suffix "-0.15" are models trained on 15% of the training pool and those with suffix "-full" are trained on the whole training data. OL, U, and M denote DialectLLM-OL, -User, and -Model, respectively.

	Turn 1			Turn 2			Turn 4			Turn 8		
Model	OL	U	M	OL	U	M	OL	U	M	OL	U	M
Qwen3-8B	0.426	0.220	0.310	0.525	0.250	0.334	0.527	0.243	0.333	0.563	0.268	0.363
Qwen3-8B-0.15	0.896	0.900	0.952	0.915	0.887	0.932	0.920	0.888	0.923	0.944	0.888	0.926
Qwen3-8B-Full	0.935	0.972	0.985	0.948	0.955	0.962	0.947	0.952	0.961	0.961	0.949	0.961
Gemma3-12B	0.468	0.275	0.396	0.459	0.267	0.439	0.458	0.264	0.484	0.507	0.285	0.542
Gemma3-12B-Full	0.946	0.978	0.988	0.953	0.963	0.971	0.950	0.958	0.967	0.962	0.954	0.967

E.5 More Results from Sec. 5.7

Table 13: Model performance on DialectLLM-Bench classification task, which are fine-tuned on different tasks. "-RC" denotes that the model is trained on the response completion (generation) task and "-Id" denotes that the model is trained on the identification (classification) task. OL, U, and M denote DialectLLM-OL, -User, and -Model, respectively.

	Turn 1			Turn 2			Turn 4			Turn 8		
Model	OL	U	M	OL	U	M	OL	U	M	OL	U	M
Qwen3-8B	0.353	0.237	0.429	0.326	0.240	0.385	0.318	0.265	0.377	0.319	0.287	0.373
Qwen3-8B-RC-0.15	0.548	0.425	0.589	0.553	0.395	0.518	0.491	0.306	0.452	0.484	0.357	0.448
Qwen3-8B-RC	0.523	0.346	0.511	0.504	0.325	0.475	0.438	0.293	0.428	0.394	0.241	0.383
Qwen3-8B-Id-0.15	0.803	0.855	0.901	0.811	0.911	0.939	0.856	0.957	0.968	0.901	0.983	0.987
Qwen3-8B-Id	0.863	0.952	0.963	0.872	0.983	0.985	0.908	0.995	0.994	0.942	0.999	0.999
Qwen3-8B-Both	0.860	0.956	0.967	0.872	0.990	0.991	0.910	0.998	0.996	0.944	0.999	0.999

Table 14: Model performance on DialectLLM-Bench generation task, which are fine-tuned on different tasks. "-RC" denotes that the model is trained on the response completion task and "-Id" denotes that the model is trained on the identification task. Models with suffix "-0.15" are models trained on 15% of the training pool. OL, U, and M denote DialectLLM-OL, -User, and -Model, respectively.

	Turn 1			Turn 2			Turn 4			Turn 8		
Model	OL	U	M	OL	U	M	OL	U	M	OL	U	M
Qwen3-8B	0.426	0.220	0.310	0.525	0.250	0.334	0.527	0.243	0.333	0.563	0.268	0.363
Qwen3-8B-RC-0.15	0.884	0.881	0.940	0.915	0.871	0.917	0.911	0.884	0.912	0.944	0.880	0.909
Qwen3-8B-RC	0.930	0.963	0.982	0.944	0.945	0.962	0.946	0.946	0.959	0.963	0.946	0.961
Qwen3-8B-Id-0.15	0.298	0.199	0.243	0.318	0.221	0.238	0.328	0.223	0.246	0.347	0.215	0.212
Qwen3-8B-Id	0.173	0.114	0.127	0.244	0.170	0.184	0.292	0.227	0.237	0.291	0.226	0.245
Qwen3-8B-Both	0.935	0.972	0.985	0.948	0.955	0.962	0.947	0.952	0.961	0.961	0.949	0.961

E.6 Impact of Inference-Time Reasoning on DialectLLM-Bench

We further investigate whether utilizing test-time compute enhances LLM performance on DialectLLM-Bench, comparing Claude-Sonnet-4 (with and without thinking mode) and DeepSeek-V3.1 versus DeepSeek-R1. As illustrated in Fig. 16, inference-time reasoning yields a substantial and consistent accuracy gain across all turns, indicating that additional computational overhead enables synthesizing subtle dialectal cues.

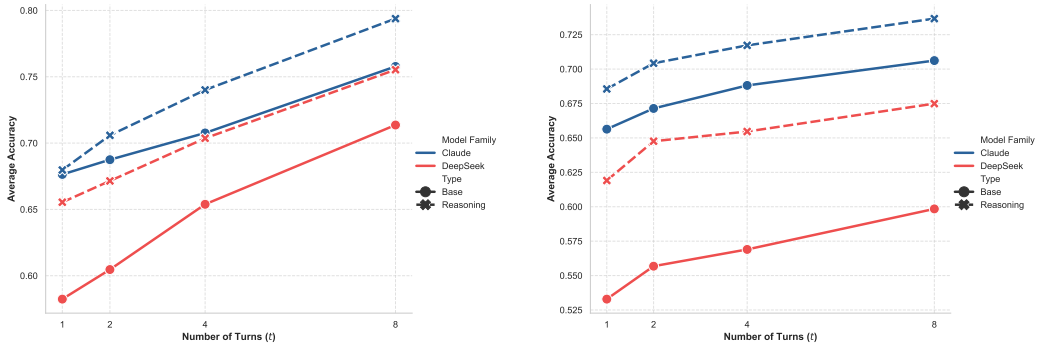


Figure 16: Impact of test-time compute on classification (left) and generation (right) task. Models equipped with reasoning capabilities (dotted lines) achieve a consistent performance increase over their base counterparts.

F Models, Database, and Hyperparameter Details

F.1 Models and Database

We run Claude-Haiku-4.5, Claude-Sonnet-4, Claude-Opus-4.1, Qwen3-32B, Qwen3-235B-A22B, Deepseek-V3.1, Deepseek-R1, GPT-OSS-20B, and GPT-OSS-120B on Amazon Bedrock. We run Qwen3-{0.6B, 1.7B, 4B, 8B, 14B} (Apache-2.0) and Gemma3-{4B, 12B, 27B} (gemma) from Huggingface on 8 H100 or 8 A100 GPUs for inference and training. We use the rules from eWAVE database [23] (CC-BY-3.0) for the morphosyntactic rules.

F.2 Hyperparameters

We present the hyperparameters used for supervised fine-tuning on classification and generation tasks.

- Learning rate = 2e-05
- Epoch = 1
- Learning Rate Scheduler= Cosine Annealing LR scheduler
- Maximum Gradient Norm = 1
- Warmup Ratio = 0.1
- Optimizer = AdamW

Moreover, for LLM inference calls on DialectLLM-Bench, we set the temperature to 0 for non-reasoning models, where supported, to control randomness. For reasoning models, where setting the temperature to 0 is not supported, we did not specify any inference hyperparameters to minimize intervention.

G Prompts

G.1 Prompt for Sec. 3.2

Seed Dialog Generation (Natural)

You are tasked with creating a natural, multi-turn conversation between a USER and an ASSISTANT about the topic: `({us_word})`

Rules:

- Generate a conversation in Standard American English (USA) with `{num_of_turns}` turns.
- If the topic `({us_word})` is related to spelling (e.g., words ending in -or), come up with an example that has the corresponding spelling structure.
- Make the conversation feel natural and conversational.
- Each turn = one USER message + one ASSISTANT response.

Format:

```
Turn 1:
USER: [message]
ASSISTANT: [response]

Turn 2:
USER: [message]
ASSISTANT: [response]
[continue as needed]
```

Figure 17: Prompt for *natural* multi-turn seed dialog generation.

Seed Dialog Generation (Indirect)

You are tasked with creating a natural, multi-turn conversation between a USER and an ASSISTANT about the topic: `[{us_word}]`

Rules:

- Generate a conversation in Standard American English (USA) with `{num_of_turns}` turns.
- If the topic `({us_word})` is related to spelling (e.g., words ending in -or), come up with an example that has the corresponding spelling structure.
- The USER must **not** directly mention the word `''{us_word}''` in the **FIRST** turn.
- In the first turn, the USER should approach the topic indirectly through:
  - Describing situations where it's needed
  - Asking about related problems or scenarios
  - Mentioning associated activities
  - Describing its features without naming it
- Make the conversation feel natural and conversational.
- The ASSISTANT can mention `''{us_word}''` in any turn. The USER can mention it starting from the **second turn**.
- Each turn = one USER message + one ASSISTANT response.

Format:

```
Turn 1:
USER: [message]
ASSISTANT: [response]

Turn 2:
USER: [message]
ASSISTANT: [response]
[continue as needed]
```

Figure 18: Prompt for *indirect* multi-turn seed dialog generation.

G.2 Prompt for Sec. 3.3

Initial Orthographic and Lexical Transformation

Convert the multi-turn dialogue from Standard American English Dialect to {target_dialect} English Dialect.

Instructions:

- Convert the dialogue while maintaining the original meaning, tone, and conversational flow
- Convert all measurements to {target_dialect} standard units
- Ensure the conversion reflects authentic linguistic features of the {target_dialect} English dialect
- Focus on genuine dialectal variations rather than exaggerated stereotypical representations
- After the conversion, provide a detailed explanation of all changes made

CONSTRAINTS:

- Do not change the original morphosyntactic features. (differences in grammar, including how words are formed (morphology) and how they are arranged in sentences (syntax).)
- You are only allowed to make lexical, orthographic, and unit transformations mentioned below.
- Keep the **number and segmentation of turns** identical to the original.

Dimensions to Consider:

- Lexical Transformation: This dimension refers to the differences in vocabulary or the specific words used to refer to the same concept
- Orthographic Transformation: This dimension involves variations in the spelling of words. The pronunciation and meaning are generally the same, but the written form differs.
- Unit conversions (e.g., standard metric vs imperial system)

The {target_dialect} English equivalent of ``{us_word}'' is ``{target_word}''.

FORMAT:

```
CONVERTED DIALOGUE:
[{target_dialect]} English version here with
grammar/prepositions/modals unchanged; same number of turns]
[Use the Turn X format with USER/ASSISTANT labels as the
original dialogue]

EXPLANATION OF CHANGES:
- List every change you made, grouped by dimension
(Orthographic, Lexical).
- Include an ``INTENTIONAL NON-CHANGES (Morphosyntax)''
section noting items you did not change because it would have
altered grammar, prepositions, or modals.
```

Original Dialogue to Convert:

{dialogue}

Figure 19: Prompt template for the construction of the initial version of the DialectLLM-OL dataset (before refinement). The italicized sentence is included only when the mapping originates from the target transformation wordbank.

G.3 Prompt for Sec 3.3

Morphosyntactic Transformation

`Previous Dialogue: {prev_dialogue}`

You are a linguistic expert specializing in {target_dialect} English dialects. Your task is to convert the text into an authentic-sounding {target_dialect} English dialect based on the given rule and then explain the specific changes you made.

Instructions:

- Refer to the previous dialogue for context understanding, do not transform the previous dialogue.
- Your primary goal is to maintain a natural flow and preserve the original meaning. Apply the transformation rule listed in the "Dialect Rule" section.
- **Pay careful attention to the human comments for specific implementation details.**
- After the conversion, provide a detailed explanation of all changes made.
- Consider the previous dialogue context to ensure consistency.
- **If the rule cannot be naturally applied, you may skip the transformation.** Apply rules for naturalness, not for every applicable candidate.

Format your response as:

```
CONVERTED DIALOGUE: [Provide the {target_dialect} version here]
EXPLANATION OF CHANGES: [Provide your explanation here]
```

Dialect Rule
Rule: {transformation_rule}
Example of original sentence: {original_sentence}
Example of transformed sentence: {transformed_sentence}
Description: {rule_description}
`Related Human's Comments: {human_comments}`

New dialogue to transform ({[user or assistant]}):
{utterance_to_transform}

Figure 20: Prompt for the construction of DialectLLM-User and DialectLLM-Model datasets.

G.4 Prompt for Sec. 3.4

Wordbank Aggregation (filling out the missing words)

Task: You are a linguist of {target_dialect} English. You will be given a Standard American {orthographic/lexical} term. Output the {target_dialect} English equivalent ONLY if it meets these criteria:

Rules:

- It is a commonly used, mainstream {target_dialect} English term (not rare, archaic, or regional)
- It is the standard/primary term used in {target_dialect} English
- **If the Standard American English term is used mostly or if no such prevalent {target_dialect} equivalent exists, output the Standard American English {orthographic/lexical} term unchanged.**

The Standard American English {orthographic/lexical} term to translate: {us_word}

Format:

```
WORD: <the {target_dialect} English equivalent or the original US term>
```

Figure 21: Prompt for filling out the missing words in the wordbank aggregation step. (See Fig. 7 for visual depiction)

Wordbank Aggregation (filling out the missing ratings)

Task: Compare the original (US English) word/spelling with the transformed ({target_dialect} English) variant on a 1-4 scale:

Rules:

- **1:** Feature is not used. The transformed word/spelling is never correct in {target_dialect} English.
- **2:** Feature exists but is rare. The word/spelling might be understood but is outdated, region-specific, or influenced by other dialects. LLM should understand this term but not generate it.
- **3:** Feature exists and is commonly used in colloquial speech. The word/spelling is widespread in informal {target_dialect} English but may coexist with other variants. Both forms are acceptable.
- **4:** Feature is required in this locale. Using the US variant would sound unnatural or incorrect. The {target_dialect} variant is the standard and expected form.

You will be given a American English version of a word/spelling and its corresponding {target_dialect} English transformation. Based on the criteria above, provide a rating from 1 to 4.

Reference Dataset:
{human annotated guideline}

Words to Evaluate:
US English: {us_word}
{target_dialect} English: {target_dialect_word}

Format:
`Rating: <number from 1 to 4>`

Figure 22: Prompt for filling out the missing ratings in the wordbank aggregation step. (See Fig. 7 for visual depiction)

Australian English OrthoLex Refinement Guideline

en-AU General Orthographic (Spelling) Conventions

- -our (e.g., colour, flavour, honour)
- -re (e.g., centre, theatre, metre)
- -ise (e.g., organise, realise)
- -yse (e.g., analyse, paralyse)
- -ce (e.g., defence, licence, offence)
- -ogue (e.g., catalogue, dialogue)
- e (e.g., encyclopedia, maneuver) OR ae and oe (e.g., encyclopaedia, manoeuvre) both acceptable
- doubled consonant (e.g., travelling, jewellery)

en-CA Specific Lexical & Orthographic Transformations

```
| Original | Transformed | Category |
| aubergine | eggplant | lexical |
| yes | nah yeah (for responsive phrases only; other cases
just stick with yes| lexical |
| tire (wheel) | tyre (wheel) | orthographic |
```

Figure 23: An exemplar guideline for the OrthoLex refinement step for Australian English. We provide the model with general spelling conventions for all transformations, while specific changes are provided through probabilistic sampling when the terms appear in the dialogue ("aubergine", "yes", "tire" in this example). These transformations are presented in a tabular format, as LLMs demonstrate superior performance when given information in structured forms [33].

G.5 Prompt for Sec. 4.1

Dialect Identification (Classification) Prompt

You are an expert linguist specializing in identifying different English dialects.

Your task is to analyze a {turn}-turn dialogue where the user's and assistant's response is provided in a certain English dialect.

The answer options are as follows:
(A) Indian English (India)
(B) Canadian English (Canada)
(C) Scottish English (Scotland)
(D) Standard American English (USA)
(E) Nigerian English (Nigeria)
(F) Irish English (Ireland)
(G) Australian English (Australia)
(H) Philippine English (Philippine)
(I) Standard British English (England)

Instructions:

- Carefully examine the morphosyntactic, lexical, and orthographic features in the dialogue.
- Based on your analysis, determine the dialect of the conversation.
- Provide your answer in the format specified below.
- Your answer should be one of the options above.
- Do not give explanations.

Required Output Format:

```
ANSWER: <Your Answer; Choose the correct option.>
```

Dialogue to Evaluate:

{dialogue}

Figure 24: Prompt template for dialect DialectLLM-Bench classification task. Note that the answer options are shuffled for each instance and when testing the DialectLLM-OL dataset, we remove the mention of "morphosyntactic" for the first instruction.

G.6 Prompt for Sec. 4.2

Dialect Response Completion (Generation) Prompt

You are an expert linguist specializing in {target_dialect]} English dialect.

Your task is to select the most appropriate response to complete a {turn}-turn dialogue. The dialogue exhibits specific dialectal features: lexical, orthographic, and morphosyntactic features.

You will be provided with:
1. The Conversation History (Conversation history of previous turns).
2. The Current User Turn (The message the Assistant must reply to).
3. A list of Candidate Options for the Assistant's response.

Instructions:
- Analyze the {sentence} of the 'Conversation History' and 'Current User Turn' to identify the specific English dialect being used.
- Evaluate the 'Candidate Options' and select the one that is linguistically consistent with the identified dialect from the conversation history and current user turn.
- Provide your answer in the format specified below.
- Your answer should be one of the options above (provide the corresponding index/label). Choose the most probable option.
- Do not give explanations.

Required Output Format:
```
ANSWER: <Your Answer; Choose the correct option; Provide the
alphabet character of the chosen option.>
```

Conversation History:
{context_history}

Current User Turn:
{current_user_turn}

Candidate Options:
{possible_options_str}

Figure 25: Prompt template for dialect DialectLLM-Bench generation task. Note that the answer options are shuffled for each instance and when testing the DialectLLM-OL dataset, we remove the mention of "morphosyntactic" features.

G.7 Prompt for Appendix D

LLM-as-a-Judge (Lexical Judge) for Open-Ended Generation Evaluation

You are an expert linguist specializing in English lexical and vocabulary variations for {target_dialect]} English dialect. Your task is to evaluate a text based STRICTLY on its lexical choices (vocabulary) for {target_dialect]} English dialect.

Evaluation Guidelines:
{user_provided_guidelines}

Binary Classification Scale:
1 - Appropriate (ALL vocabulary choices are consistent with {target_dialect]} English, or the text uses universal/neutral terms that do not violate dialect norms).
0 - Inappropriate (The text contains ONE OR MORE words or idioms that belong to a conflicting dialect).
N/A - Not Applicable (The text contains no words that have variant lexical aspects/vocabularies across dialects).

Instructions:
- The 'Evaluation Guidelines' consist of the human annotation guidelines and their corresponding annotations.
- Use the 'Evaluation Guidelines' as a reference frame to identify acceptable variations (STRICTLY FOLLOW THE GUIDELINES GIVEN).
- **SCORE CRITERIA:** To receive a score of 1, "ALL" lexical components must be appropriate. If the text uses a term from a conflicting dialect or is inappropriate, the score must be 0. If the input text does not contain any words that differ in vocabulary between dialects, output "N/A".
- Focus ONLY on word/vocabulary choices.
- Ignore spelling and grammatical structure.
- Please detail the factors contributing to this assessment.

Required Output Format:
```
Reasoning: <Explanation>
Score: <0, 1, or N/A>
```

Input Text:
{response_to_evaluate}

Figure 26: Prompt for LLMaaJ that evaluates lexical feature appropriateness.

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   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: Currently, we are going through the approval processes required to release the data and code as part of this research in accordance with our organization's guidelines, and our ability to release the artifacts is predicated on the approval decision. We have provided detailed steps on dataset generation (Section 3), along with the prompts used (Appendix G).

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   Justification: Experiment settings on evaluation and training are explained in Section 5 and Appendix F of the paper.

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